

The Impact of Ending Mandatory Union Fees: Evidence from Administrative Data*

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Abstract

The 2018 Supreme Court decision in *Janus v. AFSCME* eliminated mandatory union fees for public-sector workers, overturning four decades of legal precedent. Using administrative payroll data from 400 jurisdictions across 21 states, I find that dues-paying membership declined 8.9 to 13.4 percent by 2021—a drop smaller than anticipated and one not observed among USPS and private-sector workers. At least three-quarters of the decline reflects the automatic termination of agency fees rather than voluntary exits. Teachers, police, and firefighters maintained stable membership, while support staff declined 13 to 18 percent. Despite these membership losses, I find no impact on earnings.

Keywords: Public-sector unions; Union membership; Collective bargaining; Agency fees; *Janus*; Free-riding

JEL classification codes: J45, H75, K31, J51.

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1 Introduction

On June 27, 2018, the U.S. Supreme Court ruled in *Janus v. AFSCME* that public-sector workers could not be compelled to pay union dues or agency fees as a condition of employment. The decision, split 5–4, overturned four decades of precedent set by the Court in *Abood v. Detroit Board of Education* (1977). Under *Abood*, states were permitted to require non-union members to pay “fair share” fees to cover the costs of collective bargaining to mitigate the free-rider problem. In *Janus*, the Court held that such mandatory fees violated the First Amendment rights of public-sector employees who objected to supporting unions. This decision immediately affected approximately 6 million public-sector employees in the 21 states that had not enacted right-to-work (RTW) laws (DiSalvo, 2019). In this paper, I analyze the impact of this landmark ruling, which represents the most significant shock to public-sector labor relations in over four decades.

Observers predicted substantial effects from *Janus*.¹ Union advocates warned of devastating membership and revenue losses, with the National Education Association speculating that the decision could result in the loss of “hundreds of thousands” of members.² Despite this intense scrutiny, there has been little systematic examination of the decision’s impacts. This paper fills that gap by conducting the first academic analysis of the impact of *Janus* on dues-paying membership. Using administrative data obtained via Open Records requests from 400 state and local governments across the 21 affected non-right-to-work states, I document changes in public-sector dues-paying membership between 2018 and 2021.³ Additionally, I examine the downstream effects on public-sector earnings.

I document three key results. First, the number of employees paying union dues or agency fees declined by 8.9 to 13.4 percent from 2018 to 2021. The higher number captures membership-weighted losses, reflecting steeper declines in larger jurisdictions. Crucially, this measure captures both full union members and, in the pre-*Janus* period, agency fee payers—non-members who were required to pay roughly 85 percent of full dues to cover bargaining costs. At least three-quarters of this decline reflects the automatic termination of agency fee payments rather than voluntary exits by union members.

The decline is credibly identified across both the immediate and medium-run horizons. For the immediate response, event study analysis using monthly data confirms that these declines

¹See, for example, <https://www.nytimes.com/2018/06/27/us/teacher-unions-fallout-supreme-court-janus.html> and <https://www.npr.org/2018/06/27/606208436/supreme-court-deals-blow-to-government-unions>

²<https://www.edweek.org/teaching-learning/will-the-largest-teachers-union-lose-10-percent-of-its-members/2018/05>

³For brevity, “membership” refers to the count of employees making any financial contribution to a union, rather than formal union membership, unless otherwise noted.

began immediately *after* the Court’s decision and were not preceded by pre-trends, consistent with the ruling driving the observed membership losses; a limited comparison against right-to-work jurisdictions in Florida also points in the same direction. For the medium-run response, two within-state difference-in-differences specifications, one using USPS workers and one using private-sector workers as counterfactuals, indicate that the declines persist through 2021 and cannot be attributed to common state-level trends. Together, these three independent counterfactuals — drawn from different populations, different data sources, and different legal regimes — make it difficult to attribute the documented declines to anything other than the ruling itself.

Second, these aggregate losses mask substantial occupational heterogeneity. Using union contracts to identify employee groups covered by each bargaining unit, I find that support staff in school districts and administrative workers in municipalities and counties experienced declines of 13 to 18 percent by 2021, while teachers and uniformed personnel saw minimal losses or even gains. The common pattern linking these otherwise disparate findings is that occupations historically characterized by strong union attachment—teachers and uniformed personnel—retained members at substantially higher rates than those with weaker attachment to collective bargaining.

Third, despite sizable membership losses, I find no relationship between membership declines and subsequent earnings growth. Using payroll data from the Census of Governments, I regress changes in full-time earnings on jurisdiction-specific membership losses but fail to detect any relationship, a finding robust to state and government-type fixed effects. This null result is surprising given the large literature documenting union wage premiums, and it suggests that factors like post-pandemic labor market tightness may have offset any impact of diminished union bargaining power.

This paper contributes to the literature in three ways. First, I provide the most comprehensive evidence on the effects of *Janus* using administrative payroll records rather than survey responses, supporting causal identification through event study methods, multiple within-state counterfactuals, and heterogeneity analysis across occupations. Second, I contribute to understanding the relationship between unionization and wages in the public sector, where estimated union wage premiums typically range from 5 to 15 percent (Freeman 1986; Frandsen 2016; Brunner and Ju 2019). Third, the occupational heterogeneity in responses reveals that the institutional impact of *Janus* was concentrated, contrasting with predictions of broad organizational collapse.

The paper proceeds as follows. Section 2 provides institutional background on the *Janus* decision and public-sector collective bargaining laws, with Subsection 2.1 including a simple model that helps readers interpret the results of the paper. Section 3 describes the data collection and

sample construction. Section 4 outlines the empirical strategy for estimating the membership and earnings impacts. Section 5 reports the results, including event studies, heterogeneity analysis by occupation and government type, and earnings regressions. Finally, Section 6 concludes.

2 Institutional Landscape: The *Janus* Decision and Public Sector Collective Bargaining

While the National Labor Relations Act of 1935 governs private-sector bargaining, the rights of public-sector employees in state and local government are determined by state law. Under *Abood v. Detroit Board of Education* (1977), states were permitted to charge non-members an “agency fee” to defray the costs of collective bargaining, reasoning that non-members should not be permitted to free ride on the union’s services. Thus, public-sector employees in non-RTW states faced two options: either join the union and pay full dues, or decline membership but pay agency fees that typically amounted to 85 percent of full dues (Murphy, 2019).

On June 27, 2018, the Supreme Court overruled *Abood* in *Janus v. AFSCME*. The Court’s five Republican appointees held that mandatory agency fees constituted compelled speech in violation of the First Amendment, over the dissent of its four Democratic appointees. This decision affected public-sector employees in the 21 states that had not enacted RTW laws,⁴ relieving agency fee payers of any obligation to financially support unions and permitting union members to opt out. Importantly, *Janus* did not alter state laws requiring unions to represent all employees, including non-members, in collective bargaining, creating the potential for free-riding on the economic benefits of union representation.

2.1 A Framework for the Membership Decision

Given that unions must represent all bargaining unit employees regardless of membership status, a natural question arises as to why any rational worker would continue to pay dues voluntarily. The following framework clarifies that the membership decision depends not only on wages but also on the value of non-economic services and union attachment, and it generates testable predictions about which occupations should experience the largest and smallest membership declines.

⁴These states are: Alaska, California, Connecticut, Delaware, Hawaii, Illinois, Maine, Maryland, Massachusetts, Minnesota, Montana, New Hampshire, New Jersey, New Mexico, New York, Ohio, Oregon, Pennsylvania, Rhode Island, Vermont, and Washington. Because most other states already prohibited mandatory union payments through RTW laws, *Janus* had no direct impact there. I exclude Colorado as its Labor Peace Act required a 75 percent “super-majority” to authorize agency fees; consequently, most local and all state agencies in Colorado operated as open shops prior to 2018.

Consider a public-sector worker employed at a unionized workplace. A worker derives three types of benefits from union representation: E (economic benefits like wages), N (non-economic services such as legal representation in disciplinary proceedings), and I (consumption value from union membership encompassing labor movement solidarity, alignment with the union’s political positions, as well as peer pressure to remain in a union).⁵ Under the pre-*Janus* regime, a worker chose between paying full dues (d) for all benefits or agency fees ($f \approx 0.85d$) for E and N . A worker became a full member if the consumption value, I , exceeded the small cost differential between a union member and an agency fee-payer, i.e., if $I > 0.15d$. Post-*Janus*, the choice set changed. Non-members now paid nothing, but because many unions amended bylaws to exclude non-members from services such as legal representation, they no longer received N . Consequently, a worker now joined the union if $I + N > d$, i.e., if $I > d - N$.

This framework generates several testable predictions. The shift from a cost of $0.15d$ to $d - N$ generates aggregate declines among workers with low attachment (I) or modest valuation of union services (N). However, the impact is mediated by occupation-specific variation in N : For police, firefighters, and teachers, the high-stakes nature of professional licensure, specialized pension navigation, and due-process rights (Foy, 2025) act as a form of career insurance, keeping $d - N$ small and stabilizing membership. Support staff, in contrast, facing lower-stakes grievances and more readily available private-sector alternatives, have smaller N and thus experience larger membership losses.

3 Data and Sample Construction

3.1 Dues-Paying Membership Data from Administrative Payroll Data

I construct a panel of dues-paying membership using administrative data obtained from two sources. First, I incorporate records from the Mackinac Center for Public Policy, which used Open Records requests to collect 2018-2021 payroll data for numerous state and local governments in the states impacted by *Janus*. To construct a balanced panel, I filed supplemental requests in 2024 and 2025. For each jurisdiction, I requested: (1) the count of employees with union dues withheld (actual dues-paying members) and (2) the total number of employees covered by CBAs (regardless

⁵Although E may be endogenous to membership levels in the long run, I treat it as fixed here, motivated by the institutional reality of public-sector employment: Collective bargaining agreements typically span three to five years, and civil service protections decouple compensation from immediate fluctuations in union density.

of whether they paid union dues or agency fees pre-*Janus* or stopped paying dues post-*Janus*).⁶ Because employers in the impacted states are responsible for collecting union dues from employee paychecks and then remitting those payments to unions, employers' payroll records provide a precise census of membership status, eliminating the sampling and measurement errors prevalent in survey data like the CPS.⁷

The final sample comprises 400 jurisdictions employing 3.95 million public workers, split between 3.02 million full-time and 0.93 million part-time workers. The sample provides a comprehensive cross-section spanning all 21 affected states, including 17 state governments, 157 school districts, 123 municipalities, and 103 counties. While the median state government in the sample employs 63,540 full-time workers, the median local government employs 1,303.⁸ Furthermore, for a subset of jurisdictions, I utilize monthly payroll data from January 2018 through May 2019. These high-frequency observations allow for an event study analysis, pinpointing the precise timing of membership shifts immediately following the Court's decision—a level of granularity unavailable in survey data that is, at best, collected annually.

Separately, for the within-state difference-in-differences specifications, I utilize state-year USPS payroll records for all states between 2016 and 2023 that I obtain via a FOIA request. To construct a measure of union density, I normalize union membership by the number of employees in postal-service related occupations using the Occupational Employment and Wage Statistics (OEWS) of the BLS.⁹

3.2 Occupational Classification of Unions

To examine heterogeneity in how different types of public employees responded to the *Janus* decision, I categorize unions according to the occupations they represent. This classification is constructed by cross-referencing CBAs, union names, and jurisdictional payroll records to identify the specific employee groups represented by each union. When a union represents workers across multiple occupational categories, I assign it to the category representing a majority of its members, based on an analysis of how frequently various job titles appear in the CBA.

For school districts, I distinguish between two broad occupational categories: instructional staff (e.g., classroom teachers) and support staff (e.g., custodial or cafeteria workers, bus drivers,

⁶While these requests yielded data on dues-paying membership, data on coverage under collective bargaining were available for only about a quarter of employers.

⁷I discuss the unsuitability of the CPS at length in Appendix A.

⁸Summary statistics are provided in Table A.1.

⁹I include three occupations: 43-5051: Clerks, 43-5052: Carriers, and 43-5053: Sorters/Processors.

and administrative assistants). This distinction aligns with the "licensed" versus "classified" employee designations used by school districts and mirrors the occupational classification system established by the Census Bureau. For municipalities, I highlight four principal categories: police, firefighters, infrastructure and public works (including utility and transportation workers), and administrative and social services (including clerical staff, social workers, and general government employees). Counties and states follow a similar pattern, with corrections replacing firefighters as a major category.¹⁰

3.3 Earnings and Employment Data from Census of Governments

To measure earnings, I utilize the 2017 and 2022 Census of Governments (CoG), a quinquennial dataset covering all state and local governments. This source provides employment and payroll data disaggregated by function (e.g., K-12 education, police, and fire protection). I calculate average annual earnings for all full-time employees for 2017 and 2022 and merge these with dues-paying membership records; the final merged dataset covers all but two jurisdictions in the primary sample. These time points bracket the observation period: Fiscal year 2017 serves as the pre-treatment baseline, and fiscal year 2022 captures the medium-run post-treatment period. The merged dataset enables an examination of whether jurisdictions experiencing larger declines in dues-paying membership saw weaker earnings growth.

4 Empirical Strategy

I employ four empirical approaches to estimate the effects of the *Janus* decision on dues-paying membership and earnings.

4.1 Estimating Aggregate Membership Decline

Using annual data for 400 jurisdictions, I estimate the following panel regression to compare post-*Janus* membership levels to a 2018 baseline:

$$Y_{it} = \gamma + \sum_{t=2019}^{2021} \delta_t \cdot 1(\text{Year} = t) + \epsilon_{it}$$

The dependent variable is a within-jurisdiction index where each jurisdiction's own 2018 value

¹⁰Other occupational groups — such as higher education faculty — are present in some jurisdictions but too infrequent to support separate estimation.

is normalized to 100; thus, the coefficients δ_t represent the average percentage change relative to the pre-ruling baseline of 100. I estimate the model using both unweighted and weighted OLS, with 2018 membership as weights to capture the aggregate decline in the unionized workforce. Throughout the paper, given the small number of state clusters, I confirm inference using a wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008).

4.2 Checking for Pre-Existing Trends

For the subset of jurisdictions with monthly membership data (January 2018–May 2019), I estimate an event study to trace high-frequency membership dynamics:

$$Y_{it} = \gamma + \sum_{t \neq \text{June 2018}} \delta_t \cdot 1(\text{Month} = t) + \epsilon_{it}$$

where Y_{it} is dues-paying membership in jurisdiction i at month t , normalized to 100 in June 2018 (the reference month).

This specification assesses whether pre-existing trends could confound the results; for *Janus* to be the causal driver of membership changes, δ_t should be statistically indistinguishable from zero prior to June 2018, followed by a divergence post-ruling.

4.3 Within-State Counterfactuals

To isolate the *Janus* shock from common state-level trends, I estimate a stacked difference-in-differences specification that pools the administrative payroll sample with two within-state control groups: USPS employees, governed by the Postal Reorganization Act of 1970, and CPS private-sector employees in the same 21 states, governed by the National Labor Relations Act of 1935:

$$Y_{jt} = \alpha_j + \sum_{\tau \neq 2018} \gamma_\tau \cdot 1(\text{Year} = \tau) + \sum_{\tau \neq 2018} \beta_\tau \cdot (1(\text{Year} = \tau) \times \text{Treated}_j) + \varepsilon_{jt}$$

where α_j are unit fixed effects, γ_τ capture the control group’s counterfactual trend, and β_τ are the difference-in-differences coefficients of interest.

4.4 Cross-Sectional Regression Specification

To assess whether membership declines translated into downstream effects on earnings, I estimate cross-sectional regressions relating jurisdiction-level changes in unionization to changes in average earnings:

$$\Delta Earnings_{j,17-22} = \delta + \beta \Delta Union_{j,18-T} + X_j' \Gamma + \epsilon_j$$

The dependent variable, $\Delta Earnings_{j,17-22}$, is the percentage change in average annual earnings for full-time employees between 2017 and 2022. The regressor, $\Delta Union_{j,18-T}$, captures the percentage change in dues-paying membership from the 2018 baseline to endpoint $T \in \{2019, 2021\}$.¹¹ The vector X_j includes state and government-type fixed effects, along with baseline membership to control for heterogeneous growth trajectories.

The coefficient β represents the sensitivity of earnings growth to membership changes. A significant positive β would indicate that dues-paying membership and earnings move in tandem, such that jurisdictions with larger membership losses experience a corresponding slowdown in earnings growth—a result consistent with unions raising wages through collective bargaining.

5 Results

5.1 Dues-Paying Membership Declined Substantially Following *Janus*

Table 1 presents baseline estimates of dues-paying membership changes following *Janus*.¹² I present results from panel regressions where membership is normalized to 100 at the 2018 baseline; consequently, coefficients represent the percentage change in membership relative to this 2018 baseline. Panel A reports unweighted results to identify the average jurisdictional effect, while Panel B reports results weighted by 2018 membership to estimate the aggregate impact on the unionized workforce.

As shown in Table 1, dues-paying membership declined steadily following the ruling. Unweighted losses reached 8.9 percent by 2021 (Panel A), while membership-weighted losses were considerably higher at 13.4 percent (Panel B).^{13, 14} This gap suggests that attrition was more pronounced in larger jurisdictions, where weaker peer monitoring likely lowered the social cost of free-riding. Approximately 60 percent (unweighted) to 70 percent (weighted) of the total decline occurred within the first year, reflecting the immediate, automatic cessation of agency-fee deduc-

¹¹I winsorize earnings and membership change variables at the 5th and 95th percentiles.

¹²As defined earlier, this measure captures the total number of employees contributing financially to a union — encompassing both full members and agency fee payers. The decline in this combined measure reflects the total shock to unions; I subsequently decompose it into its automatic and voluntary components.

¹³Extrapolating these estimates to the 6 million public-sector workers affected implies a net loss of roughly 533,000–807,000 dues-paying employees nationally by 2021.

¹⁴Wild cluster bootstrap p-values are at or below 0.005, confirming the results are robust.

tions by payroll offices, followed by gradual voluntary exits by full members.¹⁵

The magnitude of these declines also varies by government type in a pattern consistent with employer size driving attrition. State governments (Column 2)—employing a median of 63,540 workers—experienced larger declines in membership, of 14.4 percent (unweighted) to 18.1 percent (weighted) by 2021. By contrast, local governments (Column 3)—employing a median of 1,303 full-time workers—experienced smaller declines of 8.6 percent (unweighted) to 10.0 percent (weighted). Among local governments, counties experienced the sharpest decline, while municipalities and school districts saw somewhat smaller declines, especially in the unweighted specifications.

To ensure these results represent a true decline in union density rather than fluctuations in the size of the workforce, I also examine union density (the ratio of members to total CoG employment, normalized to 100 at the 2018 baseline). The results, available on request, parallel the membership declines in Table 1, with density falling by 7.5 percent (unweighted) and 12.0 percent (weighted) by 2021, confirming that the observed membership losses reflect a genuine reduction in union density rather than mechanical changes in employment levels.

5.2 Identification

A possible concern with the estimates in Table 1 is that the post-2018 declines reflect confounding trends rather than the ruling itself. This concern takes two distinct forms across the two horizons I study. For the immediate response, the only plausible threat is a contemporaneous shock in the summer of 2018 that coincided with the ruling; the tight timing of *Janus* (June 27, 2018) rules out slower-moving trends by construction. For the medium-run response, the relevant threat is instead a differential trend over the 2018–2021 window that happens to track, but is not caused by, the ruling. Because these two threats differ, I address them with different evidence: monthly event-study variation and an extended pre-period panel for the immediate response, and two within-state difference-in-differences specifications for the medium-run response.

Before turning to that evidence, I can set aside one cross-cutting alternative — that a broader anti-union political shift drove the observed declines — as inconsistent with the contemporaneous political environment. The November 2018 elections produced significant Democratic gains at the state level and consequently, none of the affected states enacted RTW laws between 2018 and 2021.¹⁶ With that alternative set aside, I turn to the immediate and medium-run horizons in turn.

¹⁵These patterns hold for employers reporting coverage under CBAs, indicating minimal sample selection bias.

¹⁶States adopting RTW (e.g., Michigan) or limiting collective bargaining (e.g., Wisconsin) in the 2010s are excluded. The analysis only considers states lacking such provisions in 2018 when *Janus* was decided.

Immediate response: Figure 1 presents an event study using monthly payroll data from 36 municipalities and 17 counties between January 2018 and May 2019, with June 2018 as the omitted reference period. Coefficients for January through May 2018 are small and statistically indistinguishable from zero, indicating parallel trends in the months leading up to the decision.¹⁷ In contrast, the July 2018 coefficient indicates an immediate drop of roughly 4 percent within a single month, a shift that can be attributed to *Janus*, which was the only legal change in that window. The speed of the response also confirms its mechanical nature; because employers maintained distinct deduction codes for full members and agency fee payers (e.g., ‘1021F’ vs. ‘1021A’), payroll offices could systematically terminate all ‘A’ deductions within a single pay cycle. Table A.2 demonstrates this fact with data from three jurisdictions where agency fee deductions dropped to zero immediately following the June payrolls while full member counts remained stable or even increased.

Pre-existing trends: Extending the panel back to 2017 for a subsample of jurisdictions where such data are available allows a direct check on whether membership was already declining before *Janus*. Table A.3 reports estimates with 2017 as the reference period; across nearly all specifications, the 2018 coefficients are small and statistically indistinguishable from zero, while the 2019–2021 coefficients are large, negative, and statistically significant, suggesting membership declines of 10.8 percent (unweighted) to 18.3 percent (weighted) by 2021.¹⁸

Comparison with right-to-work jurisdictions: RTW states generally do not facilitate payroll deductions for union dues, and several have explicitly banned employers from withholding dues from paychecks. Consequently, administrative payroll data comparable to what I obtained from non-RTW jurisdictions simply do not exist for most RTW states. Florida is an exception. Using payroll data from 20 Florida jurisdictions (6 counties, 6 municipalities, and 8 school districts) collected in the first wave of data collection, I find that dues-paying membership *grew* from 118,282 members in 2018 to 121,666 in 2019 – an increase of 2.9 percent – while membership in the 400 treated jurisdictions declined by 9.4 percent. The implied DiD of 12.2 percentage points, while consistent with *Janus* driving the observed losses, should be interpreted as directional confirmation given the small comparison group.

Within-state counterfactuals: For the medium-run effect, two within-state comparisons provide additional supporting evidence. The first uses U.S. Postal Service employees, who face the same state-level political and economic conditions as state and local government workers but are gov-

¹⁷An *F*-test fails to reject the null hypothesis that all pre-*Janus* coefficients equal zero (bootstrap $p = 0.119$).

¹⁸A sole exception, state-level density in 2018, runs in the opposite direction — suggesting a slight pre-ruling *increase*.

erned by federal law and were therefore unaffected by *Janus*. Estimating the specification from Section 4.3 with USPS workers as the control group, I report the DiD coefficients in Table 2. Across specifications, my estimated declines of about 10 percent (unweighted) and 14 percent (weighted) closely track the losses in Table 1.¹⁹

The second counterfactual uses private-sector workers in the same 21 states drawn from the CPS. Although I show in Appendix A that the CPS is structurally unable to detect post-*Janus* changes among public-sector workers, the ruling did not impact private-sector workers because the First Amendment compelled-speech analysis at the heart of the ruling applied only to state actors. Thus, private-sector workers in the 21 states serve as a counterfactual. Estimating the specification from Section 4.3 with CPS private-sector workers as the control group yields a 2021 effect on dues-paying membership of roughly 9 to 10 percentage points, closely tracking the USPS-based estimates.²⁰ That two counterfactuals drawn from entirely different populations, data sources, and legal regimes converge on nearly identical magnitudes is unlikely to occur by chance. This convergence further indicates that the public-sector decline I document is not driven by state-level shocks affecting unionized workers broadly.

Taken together, these five pieces of evidence — parallel pre-trends, the extended pre-period panel, directional confirmation from Florida, and two within-state counterfactuals from USPS and private-sector workers — meaningfully address identification concerns across both time horizons and support a causal interpretation of the membership declines reported in Table 1.

5.3 Decomposing the Decline into Automatic and Voluntary Components

To assess what proportion of the overall decline is explained by the automatic loss of agency fee payers versus voluntary attrition by full members, I turn to the 46 jurisdictions that provide granular data distinguishing full union members from agency fee payers.²¹ The decomposition in Table 3 shows that the automatic elimination of agency fee payers explains virtually the entire first-year decline, with voluntary attrition by full members statistically indistinguishable from zero in 2019. By 2021, voluntary attrition reaches 4.3 percent in the unweighted specification but less than 1 percent — and statistically insignificant — with weighting. These results confirm that the initial drop following *Janus* was driven by the automatic termination of agency fee deductions. While active exits by full members were initially negligible, they grew gradually as awareness of opt-out

¹⁹If anything, the Table 2 estimates understate the true *Janus* effect: USPS density drifted modestly upward post-2018 (see Figure A.4), so differencing against USPS subtracts a positive trend from the treated trajectory.

²⁰Additional results available on request.

²¹Although representing only 12 percent of jurisdictions, they account for 57.4 percent of total baseline membership.

rights spread.

The modest voluntary attrition may appear in tension with pre-*Janus* predictions of widespread exits by full members. This resilience likely reflects proactive organizing by unions (Hodges and Malin, 2020) as well as legislative responses by states to counteract the effects of *Janus*. Fourteen of the 21 affected states adopted some combination of measures that (1) guaranteed union access to new employees, (2) restricted anti-union communications by employers and outside groups, (3) permitted unions to provide certain services exclusively to dues-paying members, and (4) limited the ways in which members could resign their membership (DiSalvo and Hartney, 2020).

To assess whether these countermeasures were effective, I interact year indicators with a binary measure of the presence of such legislation and find that jurisdictions in states with protective legislation experienced smaller overall membership declines, as shown in Figure A.1. I further decompose this pattern into its automatic and voluntary components in Table A.4. The automatic component collapses to near-zero in both groups of states, while the voluntary component shows a persistent gap favoring states with protective legislation. Thus, these laws are associated with significantly fewer voluntary exits, without any significant effect on the automatic loss of agency fee payers.

5.4 Declines in Dues-Paying Membership Concentrated among Support Staff

Having documented aggregate declines in dues-paying membership, I examine whether these losses were uniform or heterogeneous. Figure 2 illustrates these patterns across three forms of local government.²² In school districts, membership for support staff plummeted by 18 percent by 2021. This drop stands in contrast to the relative resilience of teachers' unions, which experienced a decline of only 4 percent. Municipalities confirm this divergence. While administrative and social service workers saw declines of 13 percent and infrastructure and transport workers of 8 percent, police personnel experienced essentially no change. Firefighters, in fact, saw a modest increase of about 3 percent, although that is not statistically significant at conventional levels. Counties mirror this pattern: While administrative staff and infrastructure and transport workers experienced losses of 15 and 9 percent, respectively, police membership stayed flat.²³

This cross-jurisdictional consistency, where a similar occupational ordering emerges across

²²Complete results are available in Table A.5. Weighted specifications shift individual magnitudes in both directions but preserve the same occupational ordering across all government types.

²³Figure A.2 presents occupation-specific monthly event studies for municipalities and counties, showing relatively flat pre-trends followed by an immediate drop for administrative workers, in contrast to stable membership for uniformed personnel.

three types of local government, suggests membership responses were driven by occupation-specific factors. These patterns align with the framework in Section 2.1: Occupations where the value of non-economic union services (N) is high — teachers, police, and firefighters — retained members, whereas occupations with low N saw large exits. This occupational pattern also reinforces the USPS comparison: USPS workers — predominantly clerks, carriers, and sorters — are compositionally closer to support staff and administrative workers than to teachers or uniformed personnel, suggesting that any concern about USPS workers evolving differently cuts toward *understating* rather than overstating the *Janus* effect.

A natural concern is that high-retention occupations also tend to have high attachment (I), making it difficult to separately identify the contributions of N and I . Although isolating the contribution of each channel remains challenging, the cumulative evidence offered in Appendix C favors N . In pre-*Janus* survey data, respondents cited professional security as a motive for membership over political ideology by a ratio of roughly 3:1. More directly, in 14 school districts where a single union local represents both teachers and support staff — holding political environment and union leadership constant — teachers retained members at substantially higher rates than support staff. Additionally, micro-level evidence from Wisconsin shows that teachers with higher disciplinary risk were more likely to retain membership (Foy, 2025) and local county-level measures of partisanship fail to predict the extent of membership declines — all pointing toward N , rather than I , as the key margin.

5.5 Little Evidence that Membership Declines affected Earnings Growth

Having established a significant decline in dues-paying membership, I examine whether these losses translated into slower earnings growth for public-sector employees. Table A.6 reports cross-sectional regressions relating the percentage change in average full-time annual earnings between 2017 and 2022 to the percentage change in dues-paying membership between 2018 and endpoint $T \in \{2019, 2021\}$. The results offer little evidence that membership losses affected earnings. Across both time horizons and all eight specifications, the coefficients are consistently *negative*, small in magnitude, and statistically insignificant; in the preferred specifications, the 95 percent confidence interval rules out a 10 percentage point drop in membership producing more than approximately a 1 percentage point reduction in earnings. The null result is also borne out in aggregate CoG data: nominal full-time earnings grew by approximately 15.6 percent on average between 2017 and 2022 in both *Janus*-affected and RTW states.

Occupation-specific earnings regressions reinforce this null finding.²⁴ For K–12 teachers, police, firefighters, and corrections officers, membership declines are associated with negligible earnings changes, with point estimates switching signs across specifications. Occupations with the largest declines, such as support staff and administrative workers, saw earnings growth comparable to or slightly exceeding that of occupations like police and firefighters that retained virtually all members — a pattern more consistent with pandemic-era wage compression favoring lower-paid occupations than with union bargaining power driving wage gains.

6 Discussion and Conclusions

The *Janus* ruling was the most significant shock to public-sector labor relations in over four decades, offering a rare natural experiment for examining the sources of union power. This paper exploits that experiment using a novel dataset—administrative payroll records from 400 jurisdictions across 21 states, assembled through well over a thousand Open Records requests—that captures the actual dues-paying status of millions of public-sector workers. This data effort is necessary because, as I demonstrate in Appendix A, the CPS fails to capture the effects of *Janus*, as agency fee payers never self-identified as union members prior to the ruling.

The central finding — an 8.9 to 13.4 percent decline in dues-paying membership between 2018 and 2021 — is credibly identified across both horizons. For the immediate response, a monthly event study, an extended panel starting in 2017, and a short-run comparison against right-to-work jurisdictions in Florida all point to a significant drop within a few months of the ruling, driven largely by the termination of agency fee deductions. For the medium-run response, two within-state difference-in-differences specifications — one using USPS workers and one using CPS private-sector employees as counterfactuals — indicate that the declines persist through 2021 and cannot be attributed to common state-level trends. A reader who finds any individual comparison unpersuasive might reasonably ask whether three independent counterfactuals — drawn from different populations, different data sources, and different legal regimes — could collectively point in the same direction simply by chance.

The three central findings, taken together, speak to a broader question: How do public-sector unions derive their power? The decline in dues-paying membership, while economically meaningful, was smaller than many predicted and was concentrated among support staff and administrative workers. If union influence depends primarily on the financial resources generated by

²⁴All additional results are available upon request.

mandatory fees, these membership losses could have weakened unions' ability to secure wage premiums. Instead, I fail to find a relationship between membership declines and earnings growth. This null result suggests that union bargaining power may be rooted more in the commitment of core members, institutional arrangements like exclusive representation, and the political mobilization capacity that committed unions can provide, as documented by Moe (2005), Sieg and Wang (2013), Anzia (2022), Hartney (2022), and Carreri et al. (2025).²⁵

The occupational heterogeneity of the decline provides further insight into the underlying drivers of public-sector union membership. For teachers, police, and firefighters, the value of membership extends beyond wage gains to include legal protection in disciplinary proceedings, pension advocacy, and professional identity—services that are costly to replace privately and that help unions retain members even when dues are fully voluntary. Conversely, for support staff and administrative workers with weaker historical union attachment and more readily available private-sector alternatives, the incentives to free-ride are stronger post-*Janus*.

The resilience of public-sector unions highlights a crucial political economy distinction between *Janus* and state-level RTW laws that reduced membership in teachers' unions by as much as 40 percent (Lyon, 2021). RTW laws were typically enacted as part of a wider attempt by state elected officials to curb the influence of unions and were often accompanied by restrictions such as bans on payroll deductions for union dues or requirements for annual union recertification. In contrast, *Janus* was federally imposed, primarily impacting Democratic-leaning states, 14 of which quickly enacted protective countermeasures designed to blunt its impact.

The null finding on earnings likely reflects institutional frictions that dampen the transmission of membership losses to earnings, at least over the observation window. Unions retained the privilege of exclusive representation, meaning the negotiated wage floor continued to apply to all employees regardless of membership status. Additionally, because public-sector contracts typically span three to five years, some jurisdictions were still operating under pre-*Janus* agreements. Moreover, if wage premiums adjust asymmetrically — with gains from growing membership materializing faster than losses from declining membership — then even a five-year window may be inherently underpowered to detect wage erosion.

The COVID-era labor market compounded these frictions through labor shortages and unprecedented federal fiscal aid (Clemens and Veuger, 2021). Whether the null persists as fiscal support recedes and contracts enter post-*Janus* negotiating cycles remains an open question, par-

²⁵An alternative interpretation of the null earnings result is that public-sector unions were not especially costly to employers on a per-worker basis, with any adjustment instead appearing on the employment margin—a possibility I intend to subsequently explore.

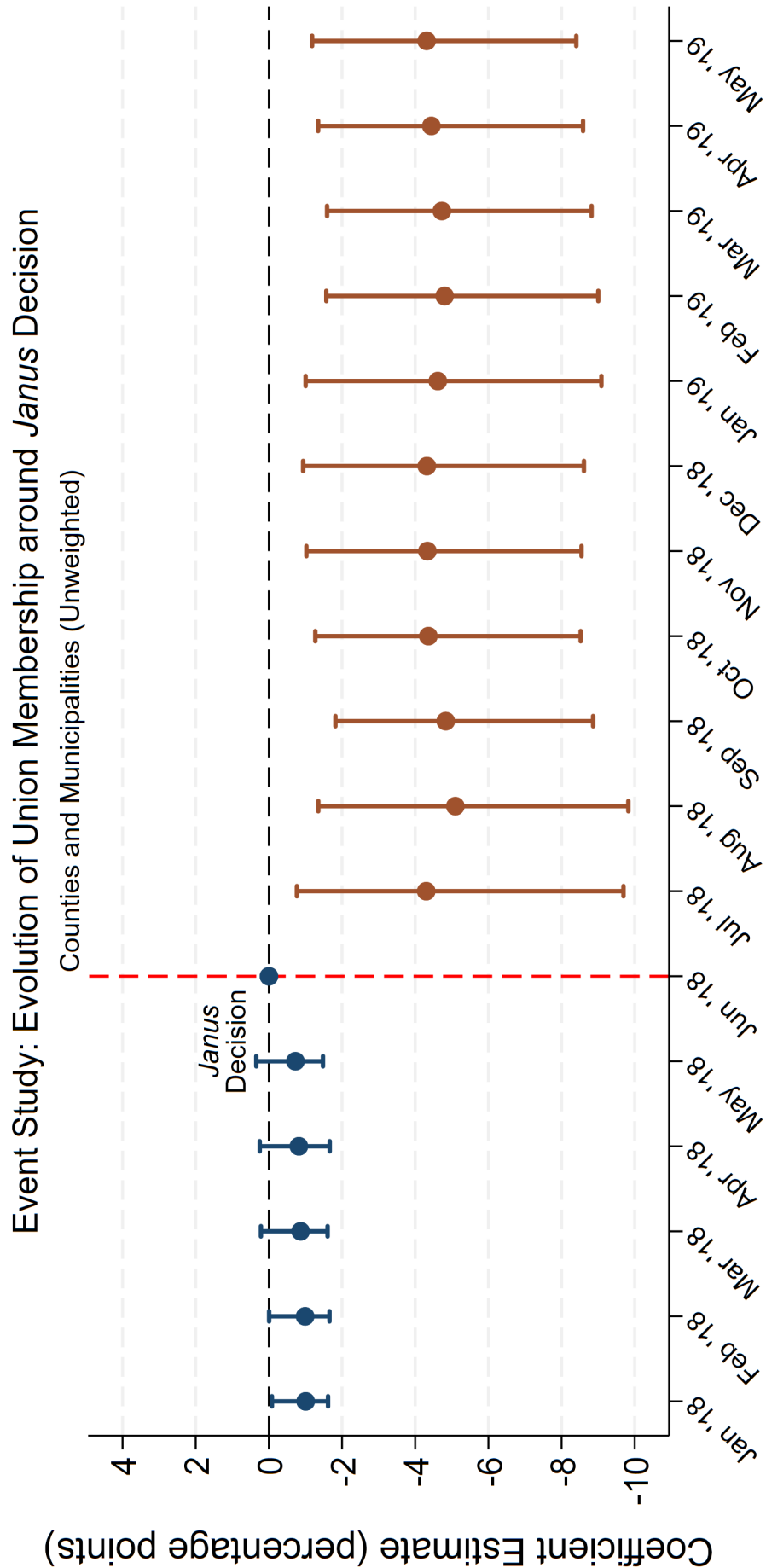
ticularly given evidence that RTW effects can take a decade to fully materialize (Lyon, 2021). Future work could identify jurisdictions where contracts were renegotiated post-*Janus* to test whether weakened unions secured less favorable terms (Biasi, 2021), or use computational contract analysis (Arold et al., 2025) to reveal whether unions shifted bargaining priorities toward member-exclusive benefits – shifts that compensation data alone cannot capture.

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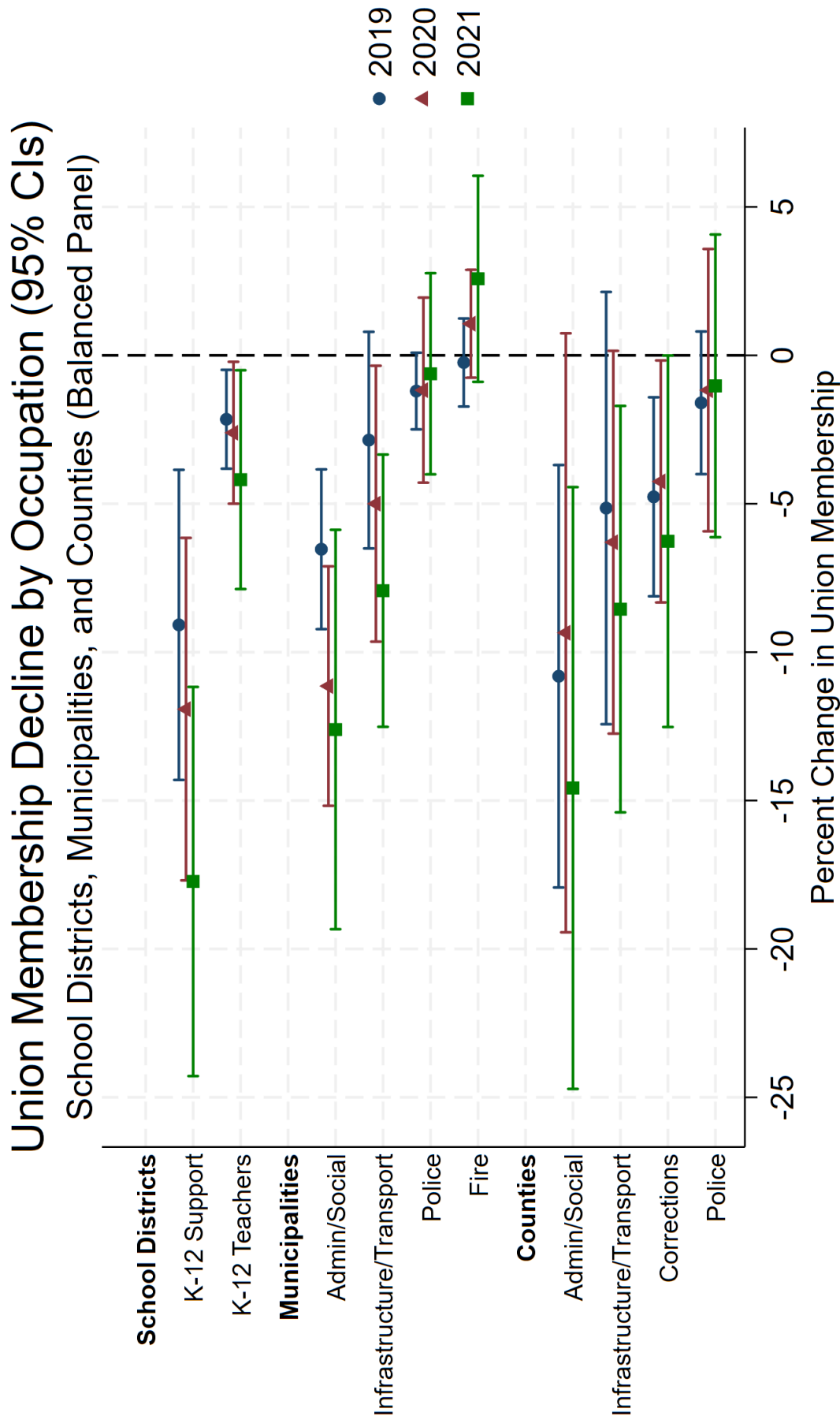
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Figure 1: Trends in dues-paying membership around the time of the *Janus* decision



Reference: June 2018 set to 0 (*Janus* decision announced June 27, 2018).
Analysis conducted on 36 municipalities and 17 counties.
95% confidence intervals from wild cluster bootstrap (Webb weights, 999 reps).
Pre-*Janus* joint test (bootstrap): $p=0.119$.

Figure 2: Occupational heterogeneity of decline in union membership



Notes: Analysis based on a balanced panel of 195 jurisdictions (89 school districts, 66 municipalities, and 40 counties) with continuous occupational data for 2018–2021. Each data point represents the estimated percent change in union membership relative to a 2018 baseline (normalized to 100). Estimates are derived via the margins command following a panel regression of membership change on period x occupation category interactions. 95% confidence intervals are shown; standard errors clustered at the state level. These results are derived from a regression specification in which observations are not weighted by baseline membership.

Table 1: Baseline estimates of the impact of *Janus* on number of dues-paying employees in non-RTW states

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable: Number of dues-paying members (normalized to 100 for the second quarter of 2018 pre- <i>Janus</i>)						
Panel A: Unweighted results for change in membership across various types of governments						
Year == 2019	-5.259*** (0.943)	-10.31*** (2.816)	-5.035*** (0.904)	-8.160*** (1.711)	-3.217*** (0.633)	-4.410*** (1.395)
Year == 2020	-7.113*** (1.058)	-11.72*** (3.033)	-6.908*** (1.015)	-11.08*** (2.183)	-5.156*** (0.666)	-5.546*** (1.642)
Year == 2021	-8.882*** (1.408)	-14.41*** (3.122)	-8.637*** (1.383)	-13.47*** (2.494)	-5.989*** (0.971)	-7.541*** (2.144)
Panel B: Weighted results for change in membership across various types of governments with 2018 membership as weights						
Year == 2019	-9.356*** (2.295)	-14.01*** (4.495)	-5.899*** (0.829)	-6.986*** (1.365)	-4.432*** (0.646)	-6.795*** (1.733)
Year == 2020	-10.86*** (2.413)	-15.17*** (4.591)	-7.655*** (0.995)	-9.106*** (1.806)	-6.924*** (0.452)	-7.724*** (2.158)
Year == 2021	-13.44*** (2.421)	-18.12*** (4.358)	-9.966*** (1.406)	-10.77*** (2.223)	-10.34*** (0.740)	-9.287*** (3.385)
Number of observations	1600	68	1532	412	492	628
Number of jurisdictions	400	17	383	103	123	157
Sample on which regressions have been estimated	Full sample	State govts.	Local govts.	Counties	Municipalities	School Districts

Notes: Standard errors clustered by state are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the number of dues-paying members, normalized to a baseline of 100 in the period immediately preceding the *Janus v. AFSCME* decision (June 27, 2018).

Data Selection and Baseline: The analysis utilizes a balanced panel of 400 jurisdictions (17 state and 383 local) with continuous annual data from 2018 through 2021. To establish a pre-treatment baseline, the 2018 index is constructed primarily from second-quarter (Q2) payroll data. For the five jurisdictions that I classify as “early implementers”—because they ceased the collection of agency fees or likely altered dues processing in anticipation of the ruling—the baseline is restricted to April and May 2018 to ensure a clean pre-decision snapshot. For the four jurisdictions where Q2 data were unavailable, Q1 2018 data were used as the baseline instead.

Weighting: Panel A presents unweighted OLS estimates. Panel B presents weighted estimates using each jurisdiction’s total 2018 dues-paying membership as weights, providing an estimate of the aggregate membership decline across the sampled population.

Jurisdictional Composition: The “Full Sample” (Column 1) includes all state and local governments. The “Local Governments” category (Column 3) comprises counties, municipalities, and school districts reported in Columns 4 through 6.

Table 2: Difference-in-Differences estimates of the impact of *Janus* on dues-paying membership and union density using USPS workers as counterfactual

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Unweighted results						
	Dep. Variable: Union Membership			Dep. Variable: Union Density		
Year == 2019	-4.460*** (0.917)	-9.513*** (2.686)	-4.236*** (0.883)	-4.660*** (0.967)	-9.001*** (2.788)	-4.467*** (0.928)
Year == 2020	-5.175*** (0.990)	-9.788*** (2.771)	-4.971*** (0.969)	-4.346*** (0.877)	-7.626*** (2.743)	-4.200*** (0.857)
Year == 2021	-10.70*** (1.432)	-16.22*** (2.813)	-10.45*** (1.435)	-9.282*** (1.225)	-12.98*** (2.875)	-9.118*** (1.219)
Panel B: Weighted results, weighted by number of employees						
	Dep. Variable: Union Membership			Dep. Variable: Union Density		
Year == 2019	-8.056*** (1.916)	-12.70*** (4.030)	-4.598*** (0.685)	-8.488*** (2.233)	-12.88** (4.637)	-5.223*** (0.729)
Year == 2020	-8.076*** (1.949)	-12.38*** (4.037)	-4.872*** (0.795)	-6.927*** (1.976)	-10.72*** (4.627)	-4.109*** (0.634)
Year == 2021	-14.15*** (2.068)	-18.83*** (3.924)	-10.68*** (1.225)	-13.76*** (2.636)	-17.68*** (5.335)	-10.84*** (1.163)
Number of observations	1684	152	1616	1684	152	1616
Sample on which regressions have been estimated	Full sample	State govts.	Local govts.	Full sample	State govts.	Local govts.

Notes: Standard errors clustered by state are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. This table reports Difference-in-Differences estimates from a stacked specification pooling the administrative payroll sample (treated) with USPS state-level observations (control). USPS employees are governed by the Postal Reorganization Act of 1970 rather than state labor statutes and were therefore legally unaffected by the *Janus* ruling.

Methodology and Model Specification: The estimates are derived from the following regression model:

$$Y_{jt} = \alpha_j + \sum_{\tau \neq 2018} \gamma_{\tau} \cdot 1(Year = \tau) + \sum_{\tau \neq 2018} \beta_{\tau} \cdot (1(Year = \tau) \times Treated_j) + \varepsilon_{jt}$$

where α_j are unit fixed effects absorbed via *reghdfe*. The year main effects (γ_{τ}) capture the USPS counterfactual trend and are not reported. The coefficients reported in the table correspond to β_{τ} , the interaction between year indicators and an indicator for state and local jurisdictions affected by *Janus* (Treated). Standard errors are clustered at the state level.

Variable Definitions: Columns (1)–(3): Normalized Membership, defined as the number of dues-paying members in jurisdiction *j* at year *t* divided by the jurisdiction's 2018 membership level, multiplied by 100. Columns (4)–(6): Normalized Density. For the administrative payroll sample, this is dues-paying membership as a ratio of total Census of Governments employment, normalized to 100 in 2018. For USPS observations, this is the percentage of the state postal workforce paying union dues, similarly normalized to 100 in 2018.

Data Selection and Baseline: The treated sample consists of the balanced panel of 400 jurisdictions (17 state and 383 local) with continuous annual data from 2018 through 2021, as described in Table 1. The control sample consists of USPS state-level observations for the 21 non-right-to-work states over the same period.

Weighting: Panel A presents unweighted estimates. Panel B presents weighted estimates using each unit's 2018 baseline membership as weights.

Jurisdictional Composition: “Full” (Columns 1 and 4) includes all jurisdictions and USPS state-level units. “State” (Columns 2 and 5) retains only state government jurisdictions alongside USPS units. “Local” (Columns 3 and 6) retains only local government jurisdictions alongside USPS units.

Table 3: Decomposition of total dues payers into automatic (agency fee payers) and voluntary (full members)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Total Payers	Agency Fees	Full Members	Total Payers	Agency Fees	Full Members	Total Payers	Agency Fees	Full Members
Panel A: Unweighted results for change in membership across various types of governments									
Year == 2019	-13.488*** (1.776)	-13.247*** (1.606)	-0.240 (0.732)	-16.782*** (3.343)	-17.490*** (2.729)	0.708 (1.330)	-12.794*** (1.800)	-12.354*** (1.663)	-0.440 (0.870)
Year == 2020	-15.988*** (2.205)	-13.411*** (1.656)	-2.577* (1.362)	-19.042*** (3.458)	-17.510*** (2.716)	-1.532 (1.959)	-15.346*** (2.384)	-12.548*** (1.711)	-2.797 (1.604)
Year == 2021	-17.723*** (2.477)	-13.417*** (1.654)	-4.306*** (1.263)	-22.187*** (3.514)	-17.510*** (2.716)	-4.677* (2.314)	-16.783*** (2.671)	-12.555*** (1.708)	-4.228** (1.417)
Panel B: Weighted results for change in membership across various types of governments with 2018 membership as weights									
Year == 2019	-12.784** (4.570)	-16.519*** (2.911)	3.734* (1.824)	-17.452** (5.962)	-19.080*** (5.220)	1.627* (0.818)	-7.612*** (2.018)	-13.681*** (1.165)	6.069*** (2.590)
Year == 2020	-14.271*** (4.540)	-16.523*** (2.913)	2.252 (1.737)	-18.907** (6.028)	-19.083*** (5.219)	0.176 (0.915)	-9.136*** (1.917)	-13.688*** (1.157)	4.552* (2.209)
Year == 2021	-17.178*** (3.962)	-16.526*** (2.912)	-0.652 (1.094)	-22.409*** (5.240)	-19.083*** (5.219)	-3.326*** (0.740)	-11.382*** (1.770)	-13.692*** (1.158)	2.310 (1.405)
Number of observations	184	184	184	32	32	32	152	152	152
Number of jurisdictions	46	46	46	8	8	8	38	38	38
Sample on which regressions have been estimated	Full sample			State governments			Local governments		

Notes: Standard errors clustered by state are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. The analysis is based on a subsample of 46 jurisdictions (8 state and 38 local) for which payroll systems separately recorded full dues payments and agency fee payments and were able to provide this information in response to Open Records Requests.

Normalization and Decomposition: Dependent variables are normalized to a total 2018 financial base (Full Members + Agency Fee Payers = 100). This additive structure ensures that the coefficients for the “Automatic” (Fee) and “Voluntary” (Member) components sum exactly to the “Total” impact observed in each triplet.

Panel and Variable Definitions: Panel A: Unweighted OLS estimates reflecting the average jurisdictional effect. Panel B: Weighted OLS estimates weighted by total 2018 financial base to reflect aggregate workforce impacts.

Total: Normalized count of all employees making any payment to the union. This can be broken up into:

- Automatic: Normalized count of agency fee payers; coefficients represent the legal elimination of compelled fees.
- Voluntary: Normalized count of full union members; coefficients represent voluntary attrition or membership conversion.

Jurisdictional Composition: The “Full Sample” (Columns 1–3) includes all governments. “State governments” (Columns 4–6) and “Local governments” (Columns 7–9) report the decomposition separately by tier. Local governments include counties, municipalities, and school districts.

Online Appendix

A Limitations of the Current Population Survey (CPS)

While much of the literature on public-sector labor trends relies on the Current Population Survey (CPS), this source is ill-equipped to capture the institutional shifts induced by the *Janus v. AF-SCME* ruling. Administrative payroll records from the jurisdictions studied in this paper reveal a 9–14 percent decline in dues-paying membership between 2018 and 2021. Yet as I show below, the CPS detects no such decline. This null result arises not because *Janus* had no impact, but because the CPS is structurally unable to measure what actually changed. Paradoxically, the CPS also often identifies union activity in jurisdictions where it is legally prohibited; for example, in 2024, 10 percent of South Carolina public-sector employees report being a union member and 13 percent report being covered by a CBA despite a complete ban on collective bargaining in the state.

1 Empirical Demonstration: CPS Shows Null Effects

To demonstrate this structural measurement gap, I analyze two variables appearing in the CPS data: the union density rate (the percentage of workers who report being in a union) and the coverage rate (the percentage of workers who report being covered by collective bargaining).

The simplest possible test of whether the CPS detects *Janus* is to compare raw means. Table A.7 reports the share of public-sector workers reporting union membership in treated and control states before and after the ruling. Between 2017 and 2021, this share fell by only 1.08 percentage points in the 21 treated states and by 0.85 percentage points in the 26 control RTW states, yielding a DiD estimate of -0.23 percentage points. Adding the private sector as a within-state benchmark produces a DDD estimate of -0.58 percentage points. Neither estimate is statistically or economically meaningful, and both stand in sharp contrast to the 8.9–13.4 percent decline documented in the administrative payroll records.

I confirm this null result more formally with different regression-based approaches, with Figure A.3 capturing the results from all three identification strategies.²⁶ Panel A presents a panel event study estimated on public-sector workers in treated states with no control group, mirroring the baseline administrative-data specification in Table 1. Panel B presents a DiD comparing treated states to pre-existing RTW states. Panel C presents a DDD that adds the private sector

²⁶Following standard practice, I exclude all 2018 observations to avoid ambiguity about treatment timing. I also drop the states of Colorado, Missouri, and Wisconsin, as the legal environment for public-sector collective bargaining was in flux in these states during this period. The estimation window spans 2015 through 2021, allowing me to trace out both the immediate post-*Janus* response and the cumulative pattern through three years after the ruling.

within those same states as an additional within-state benchmark, stripping out state-level economic shocks that might otherwise be conflated with the holding in *Janus*. Across all three panels and both outcomes, the post-*Janus* coefficients are uniformly small, and not one approaches statistical or economic significance — neither in 2019, the year immediately following the ruling, nor in 2020 or 2021. Because the panel event study in Panel A already produces a null without imposing any comparison group, the DiD and DDD nulls in Panels B and C are unlikely to reflect offsetting declines among comparison units.

These systematic null findings — across three identification strategies, two outcomes, and the full post-*Janus* period — stand in sharp contrast to the substantial decline I document in this paper using administrative payroll records sourced directly from the governments themselves.

2 Why the CPS Is Unsuitable: Three Channels

The CPS’s inability to detect the post-*Janus* membership shock reflects three distinct channels, each operating on a different segment of the affected workforce.

Channel 1: Accurate self-classification by agency fee payers.

The primary driver of the post-*Janus* decline was the cessation of mandatory deductions from agency fee payers. Crucially, these individuals already self-identified correctly as non-members in survey responses prior to the ruling. Because they never reported being union members, their transition from “fee payer” to “non-payer” does not register in the CPS as a loss of membership. The survey data are therefore accurate but conceptually blind to the administrative shock. As the decomposition in Table 3 confirms, this mechanical channel accounts for virtually the entire decline in 2019 and at least three-quarters of the cumulative decline as late as 2021.

Channel 2: Social desirability bias and cognitive misunderstanding.

Post-*Janus* survey responses may also be subject to reporting biases for two reasons. First, respondents who have ceased paying dues but remain covered by a collective bargaining agreement may be reluctant to tell a surveyor that they are free-riding — a form of social desirability bias. Second, some respondents may genuinely not understand that coverage under a CBA and active dues-paying membership are distinct statuses, leading them to report membership simply because their workplace remains unionized. Both biases would cause the CPS to overestimate post-*Janus* membership, further obscuring the modest voluntary declines that administrative records suggest. The persistent gap between survey-based and administrative measures of union membership documented in Foy (2025) suggests that such discrepancies are not merely hypothetical.

Channel 3: Baseline measurement error in union status reporting.

Finally, even for the subset of full union members who voluntarily exit post-*Janus*, the CPS may

lack the statistical power to detect the change. As Card (1996) documents, measurement error in self-reported union status is a well-known feature of the CPS, with misclassification rates of approximately 2.7 percent for membership status. A structural shift of 3–4 percent among full members — the magnitude of voluntary attrition I estimate by 2021 (Table 3, Column 3) — could easily be absorbed by this baseline noise.

Together, these three channels explain the CPS null: the first renders the largest component of the decline invisible by construction, the second introduces countervailing bias, and the third makes it likely that the remaining voluntary component falls below the survey’s detection threshold.

3 Concrete Evidence: Administrative Data from New Jersey

A concrete example illustrates this issue. For the state of New Jersey, the baseline count of financial contributors in June 2018 was 57,337, which included 10,024 agency fee payers (17.5% of the total base). By June 2019, the number of agency fee payers had fallen to zero, with the state’s response to my Open Records request explicitly citing the ruling: “Represented pay periods in 2019 and 2020 were paid after the Supreme Court decision in *Janus vs. AFSCME* eliminated agency fees”. In contrast, between 2018 and 2019, full-dues membership *grew* by about 2,400 employees.

Date	Number paying union dues	Number paying agency fees	Total number of dues payers	Number of employees covered by collective bargaining
3-Jun-16	46,105	11,139	57,244	59,891
2-Jun-17	45,943	10,927	56,870	59,707
1-Jun-18	47,313	10,024	57,337	59,998
14-Jun-19	49,700	0	49,700	59,511
12-Jun-20	48,375	0	48,375	58,833
11-Jun-21	46,729	0	46,729	60,291

Looking at this example, the net impact on the number of employees paying dues in New Jersey was a decline of about 7,600, or about 13 percent between 2018 and 2019. In the CPS framework, this 13 percent shock is absent. The 10,024 fee payers would have been (correctly) recorded as non-union members both before and after *Janus*. In fact, if the CPS had only recorded full dues-paying members, then it would have indicated an *increase* in union membership of roughly 5 percent. Likewise, over a three-year period, while I document a decline of about 19 percent in dues-paying membership for the state, the change in the number of union members is a mere 1 percent.²⁷ In fact, the CPS shows no decline between 2017 and 2021 for the state’s public-sector

²⁷Between 2018 and 2019, change in dues-paying membership = $49700/57337 - 1 = -13.3$ percent; change in union

workforce as it reports that 59.5 (59.3) percent of workers were union members in 2017 (2021).

4 Why Coverage Remains Stable

The CPS also shows no change in collective bargaining coverage, but this result is to be expected. *Janus* did not alter state exclusive representation laws; unions remain legally obligated to represent all employees in a bargaining unit regardless of dues-paying status. As the last column of the table for New Jersey captures, coverage under collective bargaining stays flat.

5 Conclusion: The Necessity of Administrative Data

The impact of *Janus* manifests itself as a decline in the number of individuals making financial contributions to unions, and the CPS fails to capture this decline. The three channels laid out above — accurate self-classification by agency fee payers, social desirability bias and cognitive misunderstanding among remaining members, and baseline measurement error — together explain why the CPS is unsuited to detecting the post-*Janus* shock. The administrative payroll data painstakingly assembled for this project thus represents not a methodological preference but a necessity for quantifying the impact of this decision.

B Theoretical Model

This Appendix fills in details of the framework summarized in Section 2.1. The simple model illuminates the choice of union membership across the two legal regimes, pre- and post-*Janus*. Consider a public-sector worker employed in a workplace covered by a collective bargaining agreement. A worker can derive utility from three sources of union-related benefits:

- *E*: Economic benefits from union representation, including wages and benefits negotiated through collective bargaining;
- *N*: Non-economic benefits from union services, such as legal representation in grievance procedures and disciplinary proceedings; and
- *I*: Consumption value from union membership that can stem from many sources such as a sense of solidarity with the labor movement, alignment with the union's political positions, the ability to vote in union elections, hold office, as well as any peer pressure from co-workers to retain union membership.

The worker faces costs that depend on the legal regime governing union security arrangements.

members = $49700 / 47313 - 1 = +5.0$ percent. Between 2018 and 2021, change in dues-paying membership = $46729 / 57337 - 1 = -18.5$ percent; change in union members = $46729 / 47313 - 1 = -1.2$ percent.

Pre-*Janus* Equilibrium Under *Abood*, states could require workers to pay agency fees to defray the costs of collective bargaining. A worker could choose between two options:

1. Be a full union member: Pay union dues d and receive utility $\mathbf{E} + \mathbf{N} + \mathbf{I}$ OR
2. Be an agency fee payer: Pay agency fees f and receive utility $\mathbf{E} + \mathbf{N}$.

A rational worker would choose full membership if and only if:

$$E + N + I - d > E + N - f.$$

Simplifying, the worker becomes a full member if: $I > d - f$. Given that agency fees typically constituted about 85 percent of full union dues, i.e., $f \approx 0.85d$, this condition reduces to: $I > 0.15d$.

In words, workers with weak attachment to unions, captured as $I < 0.15d$, chose to become agency fee payers, while other employees chose full union membership.

Post-*Janus* Equilibrium Following *Janus*, workers now choose between:

1. Be a full union member: Pay union dues d and receive utility $\mathbf{E} + \mathbf{N} + \mathbf{I}$ (as before) OR
2. Become a non-member, i.e., drop union membership entirely: Pay nothing and receive utility $\mathbf{E}$.

Importantly, while workers who opt out of membership continue to receive economic benefits E negotiated by the union (due to state labor laws requiring a union to continue to represent all workers in the bargaining unit, including non-members), they no longer receive non-economic services N . In fact, I observe that many unions amended their bylaws (and corresponding CBAs) post-*Janus* to exclude non-members from grievance processing and other services such as legal representation in disciplinary proceedings, professional liability insurance, access to union legal defense funds, and professional development programs.²⁸ To receive such legal services, workers would need to either join the union or pay for a private attorney. Therefore, faced with this new reality, a rational worker chooses membership if and only if: $E + N + I - d > E$.

This inequality simplifies to: $N + I > d$; or equivalently: $I > d - N$.

In words, workers with weak attachment to unions, captured as $I < d - N$, exit as dues payers, while other employees – with higher attachment – choose membership.

Comparative Statics and Empirical Predictions

The model generates sharp predictions about membership changes following the *Janus* decision.

Prediction 1: Aggregate membership declines

Pre-*Janus*, workers join if $I > 0.15d$. Post-*Janus*, workers join if $I > d - N$. For any $N < 0.85d$, some workers who were previously members will exit. Membership decline is weakly monotonic in

²⁸For instance, as DiSalvo and Hartney (2020) note, the New York State United Teachers union stripped non-members of the life insurance, eye and dental coverage, and other benefits it provides. Likewise, Massachusetts in 2019 changed its statutes to allow unions to charge nonmembers for the reasonable costs and fees, including arbitrator fees and related attorney's fees, for grievance or arbitration procedures arising under a collective bargaining agreement.

the change in threshold: the larger the shift from $0.15d$ to $d - N$, the greater the predicted exodus.

Prediction 2: Occupational heterogeneity in membership responses

In this simplified setup, the extent of membership decline depends critically on the value of non-economic services, N , which varies across occupations. When N is close to zero — as for clerical workers, custodial staff, and food service employees, where union grievance representation is infrequent and low-stakes — the post-*Janus* cost of remaining a member rises sharply, and many workers exit. When N is large enough to approach the full cost of dues — as for police officers and firefighters, where union representation in disciplinary proceedings can prevent termination or criminal charges, and arguably for teachers, who value tenure protection and dismissal representation (Foy, 2025) — the cost of membership barely changes, and exits are minimal. Intermediate cases, such as workers in road maintenance or municipal utilities, fall in between.

Prediction 3: Larger bargaining units experience greater membership declines

Attachment I is likely weaker in large bargaining units, where workers are geographically dispersed, social ties are attenuated, and individual membership decisions are effectively anonymous. In smaller units, peer monitoring and social sanctions may sustain membership even among workers with lower valuation of union services. If so, membership-weighted declines should exceed unweighted declines.

Prediction 4: Minimal wage effects despite membership declines

Workers who exit post-*Janus* are precisely those for whom $I < d - N$ — that is, those who place low value on union ideology and union services.²⁹ These marginal members likely contribute little to the union’s political mobilization capacity or workplace solidarity. If union bargaining power derives primarily from the commitment and engagement of high- I , high- N members — precisely those who remain in the union post-*Janus* — then the exit of marginal members may not substantially erode the union’s ability to secure wage premiums. Additionally, the model assumes E is independent of membership levels, at least in the short run. This assumption reflects several institutional features of public-sector wage-setting such as multi-year CBAs and civil service protections that establish pay scales independent of union bargaining. These features may render wages insensitive to changes in membership levels, especially in the short run.

The model thus generates the two predictions tested in the body of the paper: an aggregate decline in membership, with the largest losses concentrated in occupations where N is small.

²⁹The individual membership decision can create a collective action problem though: workers comparing $N + I - d$ to zero do not internalize the effect of their departure on future union bargaining power and the downstream effects on wages. This externality may, in fact, be particularly consequential for low-wage support staff who benefit most from union wage premiums but also have the weakest individual attachment to membership.

C Disentangling Attachment to a Union from the Value of Non-Economic Services

The framework in Section 2.1 identifies two forces sustaining membership post-*Janus*: the value of non-economic services (N) and consumption benefits of union membership (I). A fundamental empirical challenge, however, is that these two forces tend to be positively correlated across occupations. Teachers, police, and firefighters—the groups that retained members at the highest rates—plausibly both possess strong professional identities tied to collective action *and* value representation services more highly. Support staff, by contrast, likely have both lower ideological attachment and less at stake in disciplinary proceedings. This correlation makes it difficult to cleanly attribute the occupational heterogeneity documented in Section 5.4 to either channel.

In this subsection, I present four pieces of evidence that, taken together, suggest the non-economic services channel (N) plays a larger role in sustaining membership, while acknowledging that ideological attachment (I) provides a meaningful but secondary contribution.

C.1 Survey Evidence on the Motives for Membership

The 2018 “Voices from the Classroom” survey, a nationally representative survey of 1,000 teachers conducted in Spring 2018 weeks before the *Janus* decision, provides direct evidence on the relative importance of different motives for union membership. Among union members, 69.6 percent cited “professional security” as a reason for membership—the most frequently selected option by a substantial margin followed by “the union helps me professionally” (55.0 percent). In contrast, only 23.3 percent of union members cited agreement with the union’s political positions and just 14.4 percent cited agreement with the union’s non-education policy positions as to why they became members. The dominance of professional security as a stated motive—ahead of financial benefits and well ahead of political alignment—suggests that what the framework labels N (non-economic services) was the primary reason teachers valued membership, not ideological commitment to the labor movement.

The survey also asked non-union teachers—the group analogous to post-*Janus* opt-outs—why they had declined membership. The most common reason was an unwillingness to pay dues (42.2 percent), consistent with a straightforward cost-benefit calculation. Disagreement with the union’s political positions was cited by 13.9 percent, suggesting that political ideology is a meaningful but secondary factor.

C.2 Evidence from Wisconsin

Complementary evidence comes from Foy (2025), who studies the aftermath of Wisconsin’s Act 10 — legislation that, like *Janus*, eliminated mandatory union fees for public employees. In a statewide survey of teachers, 85 percent of recent union members identified “representation in case of conflict” as one of their reasons for membership, by far the most frequently cited option. Foy further documents that lower-performing teachers are more likely to be union members, consistent with workers facing higher expected costs from disciplinary proceedings deriving greater value from union representation. This pattern — teachers most exposed to disciplinary risk being the most likely to retain membership — appears in districts where unions remain legally recognized as bargaining representatives, but fades in districts where unions have lost that recognition. The Wisconsin evidence thus provides micro-level support for the mechanism the framework predicts: workers remain members primarily because of the non-economic services unions provide, not because of ideological commitment to the labor movement.

C.3 Within-Union Evidence

The cleanest evidence separating N from I comes from cases where workers in different occupations belong to the same union in the same jurisdiction, facing identical dues structures, political environments, and union leadership. The sample contains 14 school districts where (i) a single union local represents both K-12 teachers and K-12 support staff, (ii) each occupation had at least 30 dues-paying members in 2018, and (iii) total full-time employment from the Census of Governments grew by no more than 15 percent between 2017 and 2022, ensuring that within-union percent-change comparisons reflect dues-paying decisions rather than underlying workforce expansion. Across these 14 districts, teachers retained members at substantially higher rates than support staff in 11 cases, with an average within-union divergence of 14.3 percentage points (median: 12.9 percentage points).³⁰ Lowell, MA, sits near the median of this distribution: The United Teachers of Lowell represents both groups, and between 2018 and 2021 membership declined by 9.6 percent among support staff but increased by 3.3 percent among teachers.

This within-union divergence is difficult to reconcile with ideology-driven explanations, given that teachers and support staff in the same district likely hold similar political views and face similar peer environments, yet their membership responses differ sharply. The pattern instead points to occupation-specific differences in the value of non-economic services as the key mechanism.

³⁰A Wilcoxon signed-rank test rejects the null of zero divergence at $p = 0.003$.

C.4 Partisan Heterogeneity and the Limits of the Ideological Channel

If ideological attachment were the dominant force driving post-*Janus* exits, one might expect a partisan gradient: Jurisdictions in more conservative areas should experience larger membership declines as workers with lower affinity for organized labor exercised their new opt-out rights. To test this prediction, I interact post-*Janus* year indicators with county-level Democratic presidential vote share from the 2016 Presidential election, which immediately preceded *Janus*.³¹

The results, available on request, provide little support for a partisan gradient. For school districts, the interaction between the 2021 year indicator and Democratic vote share is along expected lines, i.e., positive, but statistically insignificant in both the unweighted and weighted specifications. Counties display a similar pattern. For state governments, the unweighted interaction is positive but imprecisely estimated given the small sample ($N = 17$), while the weighted estimate reverses sign. For municipalities, the weighted estimate is *negative* and statistically significant, indicating that more Democratic jurisdictions experienced *larger* declines. However, that result does not hold in the unweighted specification. An alternative specification using terciles of Democratic vote share yields similarly inconclusive results. These patterns stand in contrast to what a clean ideological attachment story would predict; if conservative workers were systematically opting out, high-Democratic-share jurisdictions would exhibit smaller declines.

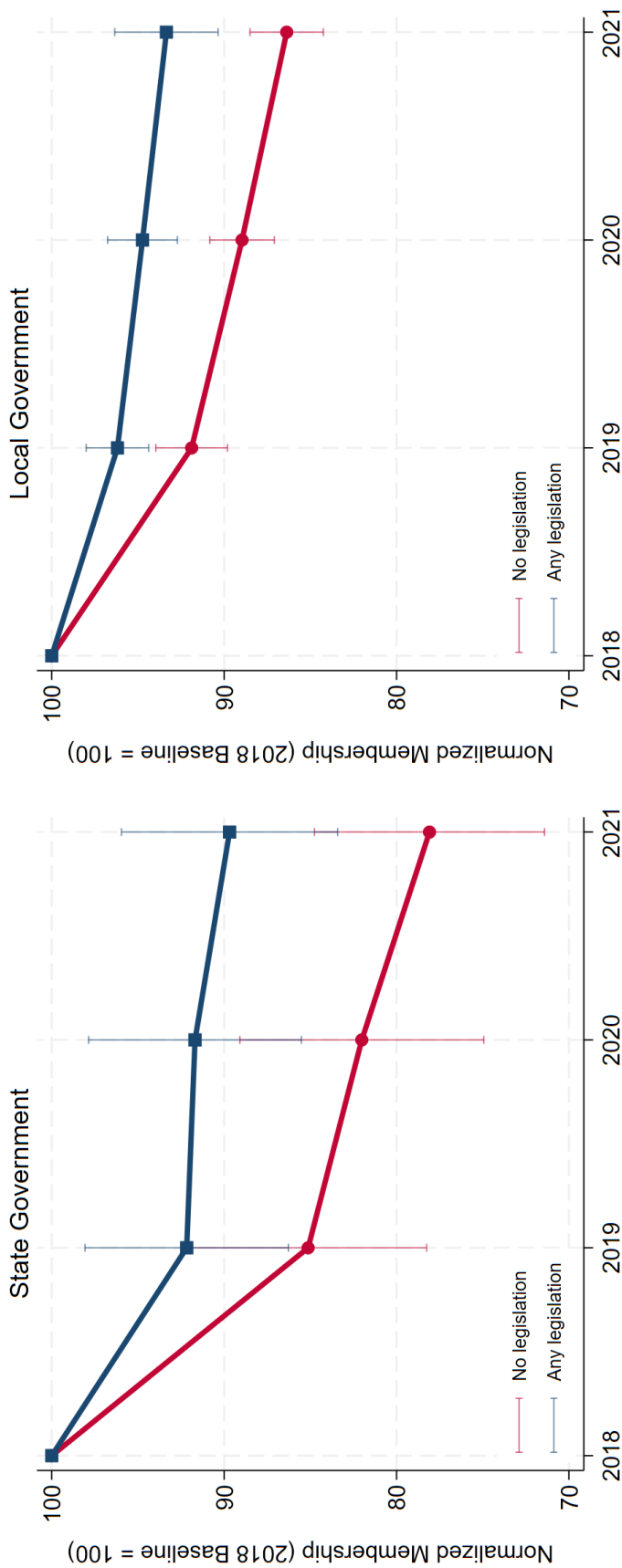
C.5 Synthesis

No single piece of evidence definitively separates N from I , but the cumulative pattern points in one direction. The pre-*Janus* survey evidence indicates that professional security dominates political ideology as a stated reason for membership by a ratio of roughly 3:1. The within-union evidence shows that teachers retain members at substantially higher rates than support staff in the same union local, in the same district, and over the same period — a divergence difficult to reconcile with ideology-driven explanations. And the legislative countermeasures analysis (Section 5.3) shows that provisions allowing unions to restrict representation services to members have a large association with reduced voluntary attrition, a finding that operates directly on N . Taken together, these results suggest that the resilience of core public-sector unions following *Janus* owes more to the concrete, excludable services unions provide — legal representation, pension navigation, due-process protections — than to workers’ ideological commitment to the labor movement.

³¹Given existing evidence that shows the passage of RTW laws depresses Democratic vote share (Feigenbaum et al. 2018), I choose to use vote shares based on the 2016 Presidential election cycle.

Figure A.1: Post-*Janus* Membership Decline by Government Level and State Legislative Protection

Post-*Janus* Membership Decline by Government Level and State Legislative Protection



90% CIs. Robust SEs for left panel; SEs clustered at state level for right panel.

Notes: Predicted margins from OLS regression of normalized membership (2018=100) on period x legislative protection x government level interactions. 90% confidence intervals shown.

Maroon line, circle markers: States with no post-*Janus* protective legislation (Alaska, Montana, Minnesota, New Hampshire, Ohio, Pennsylvania, and Vermont).

Navy line, square markers: States enacting at least one protective measure (either union access to new employees, limits on anti-union communication, allowing unions to limit services to member-only, or opt-out window restrictions). Fourteen of the 21 affected states enacted at least one measure by 2021.

Legislative coding: A state is classified as having enacted protective legislation if it passed at least one of the following types of measures by 2021: (i) mandated union access to new employee orientations or contact information (California, Connecticut, Illinois, Maine, Maryland, Massachusetts, New Jersey, New Mexico, New York, Oregon, Washington); (ii) restricted employers or outside groups from discouraging union membership (California, Connecticut, Illinois, New Jersey, New York, Oregon, Washington); (iii) authorized unions to limit grievance representation or other services to dues-paying members (Massachusetts, New York, Rhode Island); or (iv) imposed restrictions on when or how employees could revoke dues authorization, such as annual opt-out windows or union-directed revocation procedures (California, Connecticut, Delaware, Hawaii, Illinois, Massachusetts, New Jersey, New Mexico, New York, Oregon, Rhode Island, Washington). Classification draws on DiSalvo (2019), DiSalvo and Hartney (2020), state legislative databases, and review of individual statutes.

Figure A.2: Trends in dues-paying membership around the time of the *Janus* decision – Decomposed by occupation

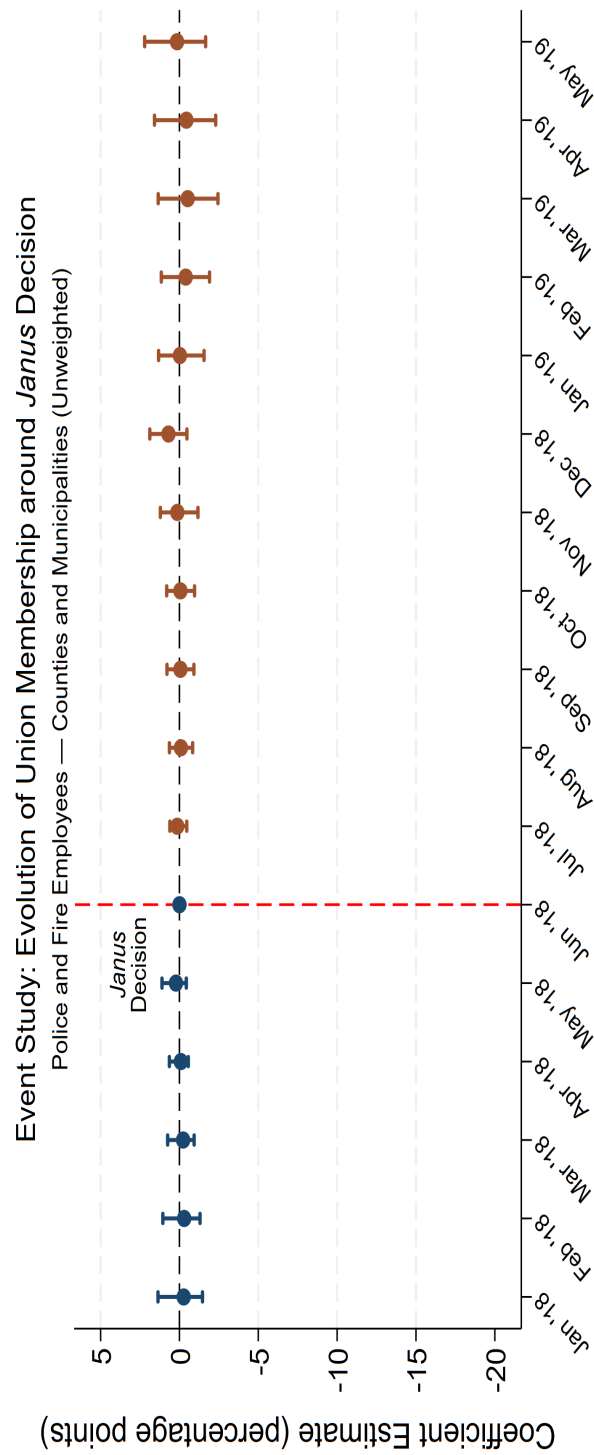
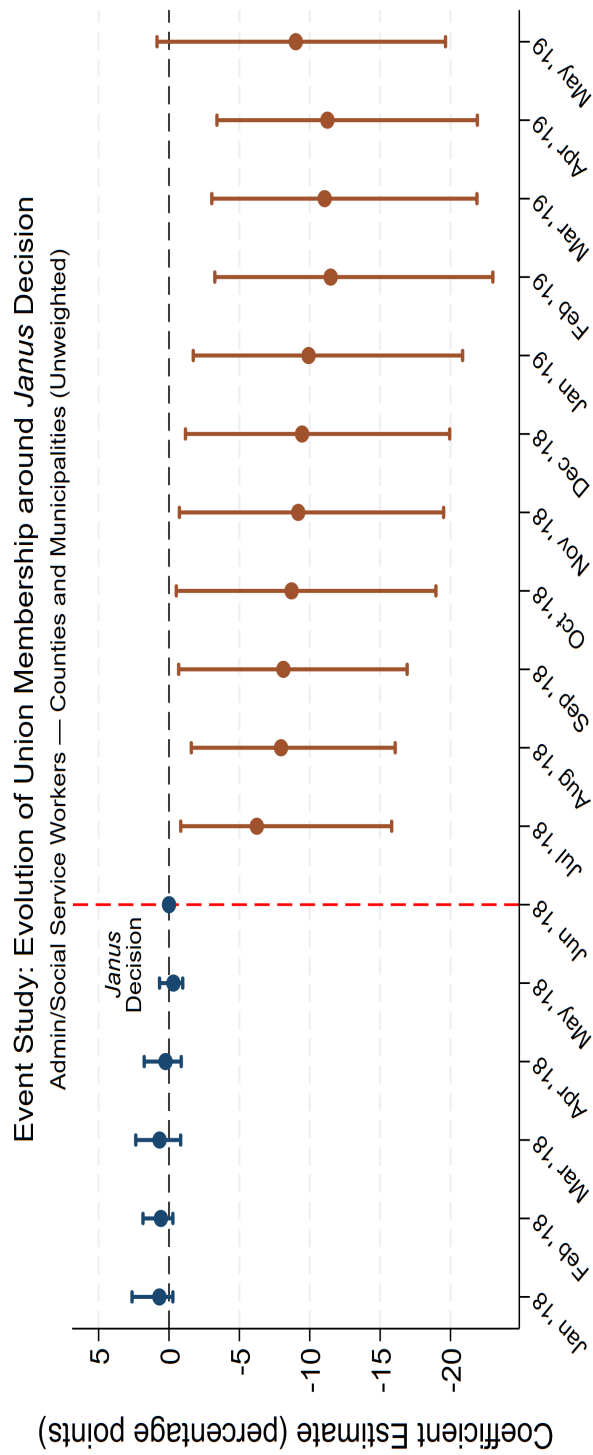
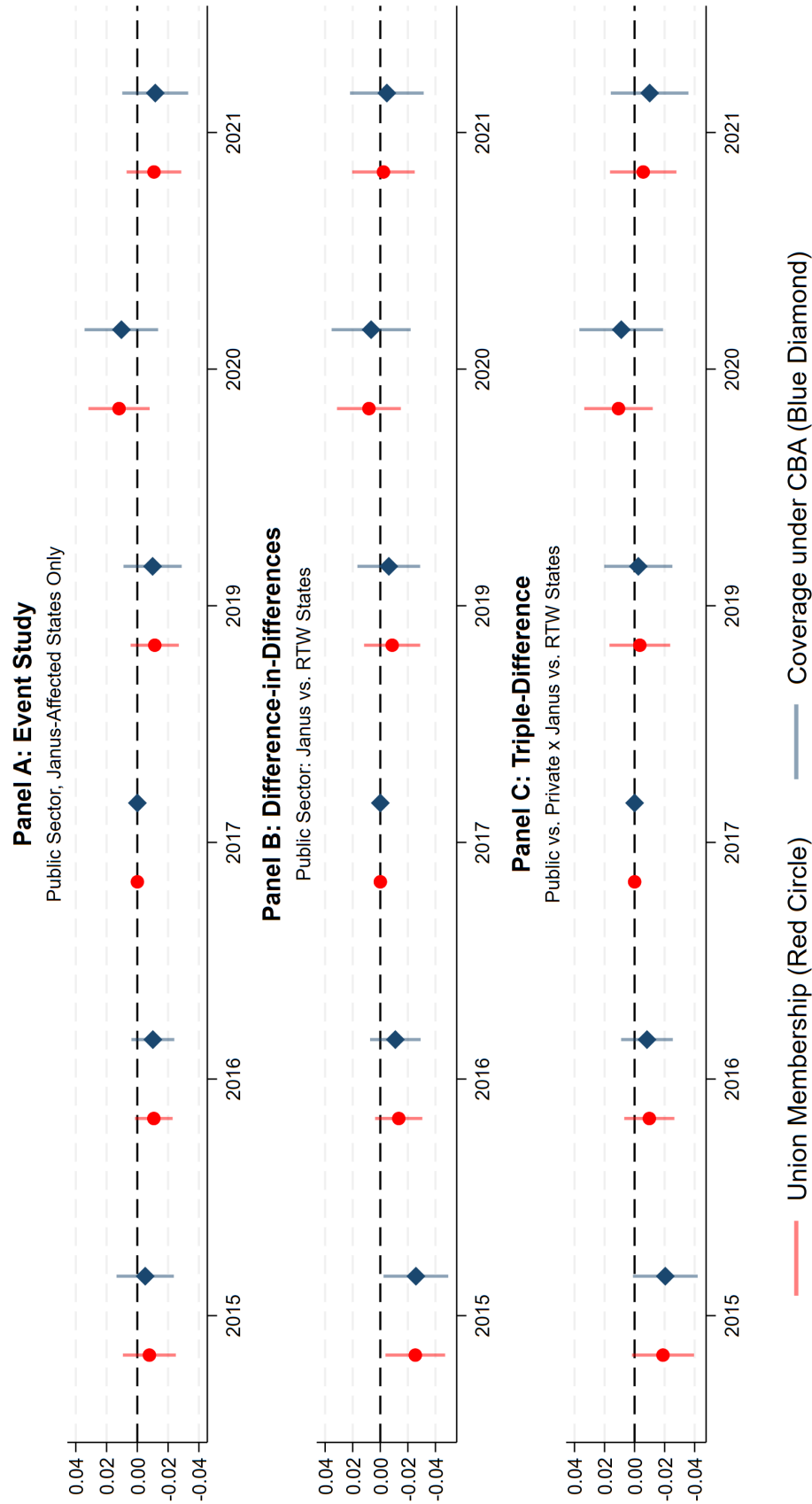


Figure A.3: Estimated impact of *Janus* on number of union members and number of employees covered by CBAs in non-RTW states using data from the CPS



2017 is the reference year. 2018 excluded. 95% CIs, clustered by state.
Y-axis: Percentage point change in state-level membership/coverage rate

Notes: This figure presents annual event study coefficients and 95% confidence intervals derived from the CPS (2015–2021) using raw state-level membership and coverage rates. Standard errors are clustered at the state level. Panel A clusters over N=21 treated states; Panels B and C cluster over 47 states.

Panel A displays the event study for *Janus*-affected states only, with no control group, mirroring the baseline administrative-data specification in Table 1.

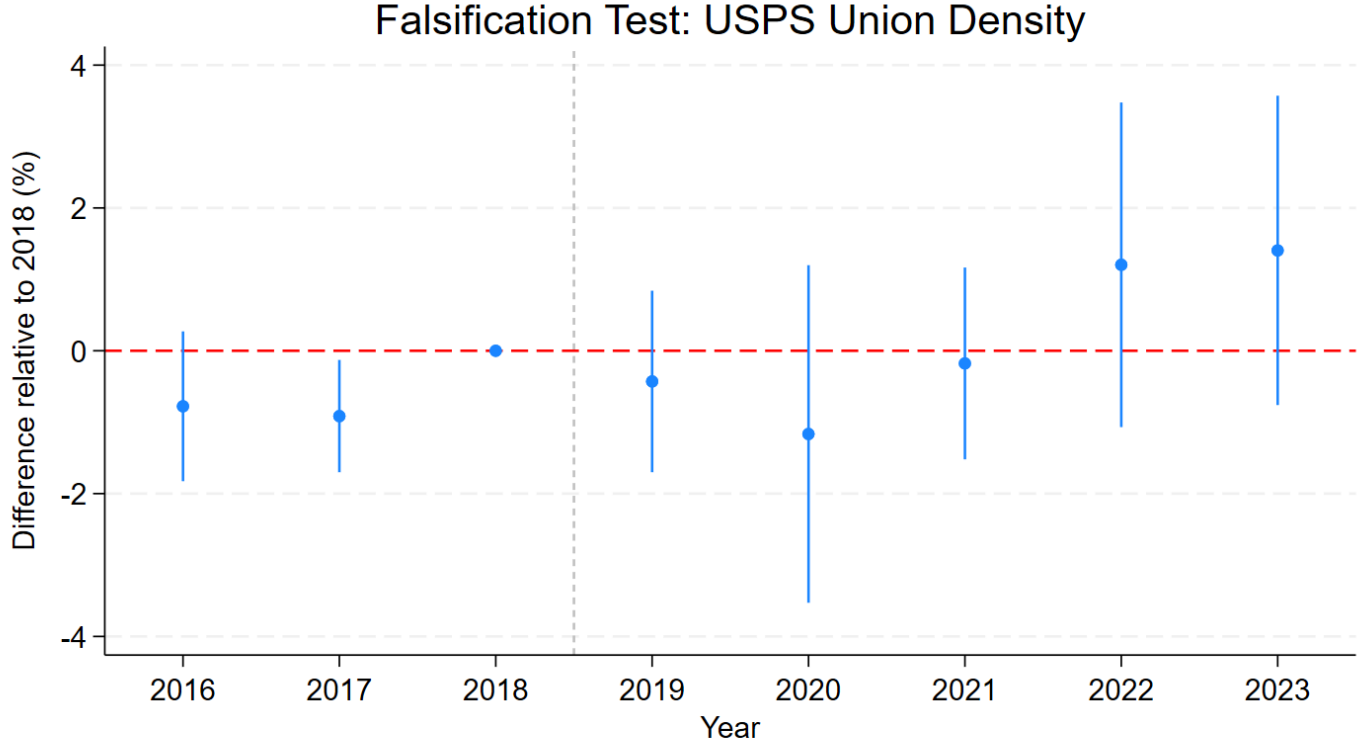
Panel B displays DiD estimates for the public sector, comparing treated states to pre-existing RTW states.

Panel C displays DDD estimates comparing public and private sectors across treated and control states.

Red Circles indicate self-reported union membership; **Blue Diamonds** indicate collective bargaining coverage.

The year 2017 is the omitted reference category; 2018 is excluded because the CPS cannot distinguish pre- from post-*Janus* respondents within that calendar year.

Figure A.4: Event Study of USPS Union Density (Falsification Test)



Note: 2018 is the reference year. Weights based on 2018 membership levels.

Notes: This figure displays point estimates and 95% confidence intervals from a weighted dynamic Difference-in-Differences (event study) specification. The sample consists of U.S. Postal Service (USPS) employees from 2016 to 2023, a group governed by federal law and legally unaffected by the *Janus v. AFSCME* decision.

Methodology and Model Specification: The plotted coefficients represent the parameters β_τ from the following regression:

$$Y_{st} = \alpha_s + \gamma_t + \sum_{\tau \neq 2018} \beta_\tau (Year_\tau \times Treat1_s) + \epsilon_{st}$$

The independent variables are interactions between year dummies and an indicator for the 21 states that allowed mandatory agency fees prior to 2018 (*Treat1*). The model includes State Fixed Effects (α_s) and Year Fixed Effects (γ_t). Standard errors are clustered at the state level ($N = 47$). The year 2018 serves as the reference period, with its coefficient normalized to zero.

Variable Definition and Weighting: The dependent variable is Union Density, defined as the percentage of the total USPS workforce in a state that pays union dues: $(Members_{st}/Headcount_{st}) \times 100$. Regressions are weighted by the 2018 baseline union membership level in each state to account for the relative size of the state-level postal workforces.

Treatment Definition: Includes the 21 non-right-to-work states where the *Janus* decision invalidated mandatory agency fees; compared with 26 RTW states, thus excluding Colorado, Missouri, and Wisconsin, from both the treatment and control groups.

Table A.1: Summary Statistics for the Estimation Sample

	Full Sample	State	County	Municipality	School District
Panel A: Sample					
Number of observations	1600	68	412	492	628
Number of jurisdictions	400	17	103	123	157
Number of states	21	17	16	20	20
Panel B: 2018 Baseline					
Dues-paying members, mean	5,311	53,294	2,105	3,900	3,323
Dues-paying members, median	889	40,186	743	429	1,846
FT employment (CoG), mean	7,554	91,120	3,388	5,026	3,218
FT employment (CoG), median	1,423	63,540	1,791	576	1,874
Union density (%), mean	59.3	39.2	45.6	56	73.1
Union density (%), median	64.1	36.9	44.8	61.9	77.6
Panel C: Decomposition Subsample					
Jurisdictions with fee/member split	46	8	13	10	15
Agency fee share (%), mean	13.5	17.5	18.6	7.7	10.7
Agency fee share (%), median	10.3	18.5	17	3.9	7.9
Panel D: Annual Earnings (CoG)					
Full-Time earnings 2017 (\$), mean	67,288	68,825	63,485	73,787	64,488
Full-Time earnings 2017 (\$), median	65,922	67,627	62,127	73,776	64,092
Full-Time earnings 2022 (\$), mean	77,881	79,644	73,928	85,316	74,415
Full-Time earnings 2022 (\$), median	76,055	79,464	73,246	82,455	72,177
Number of observations	398	17	103	123	155

Notes: This table reports sample composition and baseline characteristics for the 400 jurisdictions used in the primary analysis. All statistics are computed on a cross-section of unique jurisdictions at the 2018 baseline unless otherwise noted.

Panel A reports the number of jurisdictions and the number of distinct states represented in each government-type column. The full sample comprises 17 state governments and 383 local governments (103 counties, 123 municipalities, and 157 school districts) spanning the 21 states affected by the *Janus* decision (Colorado is excluded for reasons discussed in footnote 4). The number of jurisdictions per state ranges from 1 (Hawaii) to 54 (California).

Panel B reports baseline characteristics measured in 2018. “Dues-paying members” is the count of employees with union dues or agency fee deductions on their payroll records in the second quarter of 2018, which serves as the denominator for the normalized membership index (baseline = 100) used in Table 1. Full-time employment is drawn from the 2017 Census of Governments (CoG). The union coverage rate is defined as the ratio of dues-paying members to total employment (full-time plus part-time) from the CoG, expressed as a percentage.

Panel C restricts attention to the 46 jurisdictions (8 states, 13 counties, 10 municipalities, and 15 school districts) whose payroll systems separately recorded full union dues and agency fee payments, enabling the decomposition reported in Table 3. The agency fee share is the number of agency fee payers as a percentage of total financial contributors (full members plus agency fee payers) at the 2018 baseline.

Panel D reports average annual earnings for 398 jurisdictions from the 2017 and 2022 Census of Governments. Newark Public Schools (2017) and Providence School District (2022) are excluded due to missing earnings data during their periods of state receivership.

Table A.2: Speed of payroll adjustment following the *Janus* decision (June 27, 2018)

		May 2018	Jun 2018	Jul 2018	Aug 2018	Change Jun → Jul
Panel A: City of San Diego, California						
AFSCME Local 127 representing Employees in the Maintenance, Labor, Skilled Trades and Equipment Operator Unit	Dues-paying members	1,662	1,696	1,708	1,692	+12
	Agency fee payers	157	142	0	0	-142
	Total payers	1,819	1,838	1,708	1,692	-130 (-7.1%)
Municipal Employees Association (MEA) representing all employees in the Administrative and Field Support functions	Dues-paying members	4,235	4,275	4,284	4,258	9
	Agency fee payers	194	181	0	0	-181
	Total payers	4,429	4,456	4,284	4,258	-172 (-3.9%)
Panel B: Allegheny County, Pennsylvania						
AFSCME, Council 84 representing Professional Employees	Dues-paying members	313	315	312	306	-3
	Agency fee payers	155	163	0	0	-163
	Total payers	468	478	312	306	-166 (-34.7%)
Allegheny County Prison Employees Independent Union representing Correctional Officers	Dues-paying members	354	361	360	386	-1
	Agency fee payers	47	48	0	0	-48
	Total payers	401	409	360	386	-49 (-12.0%)
LIUNA representing workers in construction, maintenance, and public service	Dues-paying members	138	144	143	143	-1
	Agency fee payers	21	21	0	0	-21
	Total payers	159	165	143	143	-22 (-13.3%)
Panel C: Delaware County, Pennsylvania						
AFSCME Local 3107 representing employees of row offices/judicial staff	Dues-paying members	375	377	374	385	-3
	Agency fee payers	268	258	0	0	-258
	Total payers	643	635	374	385	-261 (-41.1%)
PASNAP representing Health Professionals	Dues-paying members	108	106	116	114	+10
	Agency fee payers	26	28	0	0	-28
	Total payers	134	134	116	114	-18 (-13.4%)

Notes: This table demonstrates the immediate response of public-sector payroll departments to the Supreme Court's *Janus v. AFSCME* decision on June 27, 2018. The decision eliminated mandatory agency fees for public sector workers who chose not to join unions. Data show that agency fee deductions ceased in July 2018 (the first full month following the decision), indicating that payroll departments across jurisdictions—from the City of San Diego, in California to Allegheny and Delaware counties in Pennsylvania—implemented the ruling within 2–4 weeks.

Source: Responses from respective jurisdictions under each state's Open Records Acts.

Table A.3: Estimates of the impact of *Janus* on number of dues-paying employees in non-RTW states, when data for 2017 is present

(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Unweighted results					
	Dep. Variable: Union Membership	Dep. Variable: Union Density			
Year == 2018	-0.254 (0.757)	0.158 (0.884)	-0.338 (0.910)	0.0592 (0.767)	1.647* (0.814)
Year == 2019	-7.759*** (1.661)	-11.89*** (3.492)	-6.915*** (1.619)	-6.955*** (1.767)	-8.979** (3.762)
Year == 2020	-8.799*** (1.387)	-12.99*** (3.620)	-7.943*** (1.230)	-7.410*** (1.711)	-8.627* (3.934)
Year == 2021	-10.82*** (3.446)	-15.73*** (3.477)	-9.818** (3.809)	-8.483** (3.945)	-9.921** (3.934)
Panel B: Weighted results for change in membership with 2018 membership as weights					
Year == 2018	0.00234 (0.431)	0.170 (0.641)	-0.377 (0.999)	0.288 (0.727)	0.555 (1.206)
Year == 2019	-14.75*** (4.359)	-17.54** (5.900)	-8.447*** (1.749)	-14.11** (5.012)	-16.69** (7.144)
Year == 2020	-15.76*** (4.716)	-18.62** (6.067)	-9.287*** (2.480)	-14.74** (5.636)	-17.27* (7.863)
Year == 2021	-18.34*** (5.000)	-21.17*** (5.990)	-11.92*** (3.554)	-16.91** (6.190)	-19.32** (8.338)
Number of observations	295	50	245	295	50
Number of jurisdictions	59	10	49	59	10
Sample on which regressions have been estimated	Full sample	State govts.	Local govts.	Full sample	State govts.
					Local govts.

Notes: Standard errors clustered by state are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. The analysis is based on a balanced panel of 59 jurisdictions (10 state and 49 local) with continuous annual data for the 2017–2021 period.

Normalization and Baseline: Dependent variables are normalized to a baseline of 100 representing the period immediately preceding the *Janus v. AFSCME* decision (June 27, 2018). The 2018 baseline is constructed primarily from Q2 payroll data. For “early implementers,” the baseline is restricted to April and May 2018 to exclude anticipatory changes following the oral arguments or state-level legislative shifts. For jurisdictions where Q2 data were unavailable, Q1 2018 data were utilized instead.

Panel and Variable Definitions: Panel A: Unweighted OLS estimates reflecting the average jurisdictional effect. Panel B: Weighted OLS estimates weighted by total 2018 financial base to reflect aggregate workforce impacts.

Jurisdictional Composition: The “Full Sample” (Columns 1 and 4) includes all state and local governments. The “Local governments” category (Columns 3 and 6) includes counties, municipalities, and school districts.

Table A.4: Decomposition of Membership Decline by State Legislative Counter-measures Split into Automatic (agency fee payers) and Voluntary (full members)

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Payers	Agency Fees	Full Members	Total Payers	Agency Fees	Full Members
Panel A: Unweighted results for change in membership						
Year == 2019	-17.038*** (2.655)	-15.541*** (2.151)	-1.496** (0.656)	-16.390*** (2.789)	-14.899*** (1.993)	-1.492 (0.943)
Year == 2020	-22.289*** (2.583)	-15.975*** (2.336)	-6.313*** (0.606)	-22.254*** (2.689)	-15.445*** (2.224)	-6.809*** (0.644)
Year == 2021	-23.779*** (3.158)	-15.975*** (2.336)	-7.804*** (1.304)	-23.270*** (3.363)	-15.445*** (2.224)	-7.824*** (1.373)
Year == 2019 X State enacted any legislation	5.832 (3.417)	3.769 (3.190)	2.064* (1.064)	5.694 (3.443)	4.029 (3.177)	1.666 (1.386)
Year == 2020 X State enacted any legislation	10.350*** (3.405)	4.213 (3.325)	6.138*** (1.233)	10.939*** (3.444)	4.587 (3.335)	6.352*** (1.439)
Year == 2021 X State enacted any legislation	9.949** (4.489)	4.203 (3.322)	5.746*** (1.870)	10.270* (4.743)	4.576 (3.332)	5.694** (2.041)
Panel B: Weighted results for change in membership with 2018 membership as weights						
Year == 2019	-16.350*** (1.379)	-17.233*** (1.926)	0.882 (1.129)	-10.417*** (1.557)	-12.215*** (1.701)	1.799* (0.853)
Year == 2020	-18.780*** (1.373)	-17.305*** (1.911)	-1.475 (2.251)	-13.572*** (2.510)	-12.395*** (1.690)	-1.177 (2.742)
Year == 2021	-20.912*** (1.525)	-17.305*** (1.911)	-3.607 (2.335)	-14.928*** (1.858)	-12.395*** (1.690)	-2.533 (1.470)
Year == 2019 X State enacted any legislation	4.017 (5.319)	0.805 (3.821)	3.212 (2.277)	3.028 (2.632)	-1.583 (2.091)	4.611 (2.798)
Year == 2020 X State enacted any legislation	5.078 (5.283)	0.880 (3.816)	4.198 (2.923)	4.790 (3.211)	-1.396 (2.080)	6.186 (3.516)
Year == 2021 X State enacted any legislation	4.206 (4.739)	0.877 (3.815)	3.328 (2.633)	3.830 (2.695)	-1.401 (2.081)	5.231** (2.047)
Number of observations	184	184	184	152	152	152
Number of jurisdictions	46	46	46	38	38	38
Sample on which regressions have been estimated	Full sample			Local governments		

Notes: Standard errors clustered by state are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. The analysis is based on a subsample of 46 jurisdictions (8 state and 38 local) for which payroll systems separately recorded full dues payments and agency fee payments.

Normalization and Decomposition: Dependent variables are normalized to a total 2018 financial base (Full Members + Agency Fee Payers = 100). This additive structure ensures that the coefficients for the “Automatic” (Fee) and “Voluntary” (Member) components sum exactly to the “Total” impact observed in each triplet.

Panel and Variable Definitions: Panel A: Unweighted OLS estimates reflecting the average jurisdictional effect. Panel B: Weighted OLS estimates weighted by total 2018 financial base to reflect aggregate workforce impacts.

Total: Normalized count of all employees making any payment to the union. This can be broken up into:

- Automatic: Normalized count of agency fee payers; coefficients represent the legal elimination of compelled fees.
- Voluntary: Normalized count of full union members; coefficients represent voluntary attrition or membership conversion.

Jurisdictional Composition: The “Full Sample” (Columns 1–3) includes all governments in the forensic subsample. “Local governments” (Columns 4–6) report the decomposition separately for local governments that include counties, municipalities, and school districts.

Table A.5: Occupational heterogeneity in dues-paying membership changes following *Janus*

Row Label	(1)	(2)	(3)	(4)	(5)
Panel A: School Districts	K-12 Support	K-12 Teachers			Overall
2019	-9.08*** (2.67)	-2.15** (0.85)			-5.58*** (1.64)
2020	-11.92*** (2.94)	-2.61** (1.22)			-7.21*** (1.93)
2021	-17.73*** (3.34)	-4.19** (1.88)			-10.88*** (2.48)
Observations (N)	688	688			688
Jurisdictions (J)	89	89			89
Panel B: Municipalities	Admin/Social	Infra/Transport	Police	Fire	Overall
2019	-6.53*** (1.37)	-2.86 (1.86)	-1.20* (0.66)	-0.24 (0.76)	-2.67*** (0.55)
2020	-11.14*** (2.06)	-5.00** (2.37)	-1.17 (1.59)	1.07 (0.93)	-3.95*** (0.81)
2021	-12.61*** (3.43)	-7.93*** (2.34)	-0.62 (1.73)	2.58 (1.77)	-4.39*** (1.33)
Observations (N)	848	848	848	848	848
Jurisdictions (J)	66	66	66	66	66
Panel C: Counties	Admin/Social	Infra/Transport	Corrections	Police	Overall
2019	-10.81*** (3.63)	-5.15 (3.71)	-4.77*** (1.71)	-1.60 (1.23)	-5.65*** (1.58)
2020	-9.35* (5.15)	-6.30* (3.29)	-4.25** (2.08)	-1.17 (2.43)	-5.26*** (1.81)
2021	-14.58*** (5.17)	-8.55** (3.49)	-6.27** (3.19)	-1.03 (2.60)	-7.62*** (2.24)
Observations (N)	476	476	476	476	476
Jurisdictions (J)	40	40	40	40	40
Panel D: States	Admin/Social	Infra/Transport	Corrections	Police	Overall
2019	-13.30*** (4.44)	-1.23 (10.67)	0.83 (3.21)	-0.60 (2.29)	-4.15* (2.29)
2020	-14.06*** (4.53)	-0.36 (10.71)	-0.11 (2.79)	1.28 (4.31)	-3.99 (2.48)
2021	-16.59*** (4.15)	-4.09 (8.69)	-3.93 (2.61)	-2.93 (4.28)	-7.52*** (2.20)
Observations (N)	132	132	132	132	132
Jurisdictions (J)	10	10	10	10	10

Notes: Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are reported in parentheses below the coefficients and are clustered by state for all local governments – counties, municipalities, and school districts – and are robust to heteroskedasticity for state governments.

This table reports estimated marginal effects representing the total percent change in dues-paying membership relative to the 2018 baseline (normalized to 100). The analysis is based on 205 jurisdictions (89 school districts, 66 municipalities, 40 counties, and 10 states) with complete occupational data for 2018–2021.

Specifications: Results are shown for the unweighted, balanced panel.

Occupational Groupings: School districts include K-12 support staff and teachers. Municipalities include administrative/social service workers, infrastructure/transport, police, and firefighters. Counties and states each include administrative/social service workers, infrastructure/transport, corrections, and police. The overall column reports the average across the occupation categories shown in columns (1)–(4). There are occupational categories that are too sporadically present that are not included in the estimation here but are included in the baseline results in Table 1, e.g., healthcare workers employed by municipal governments or firefighters in state governments (e.g., CAL FIRE Local 2881).

Ordering: Columns are ordered from left to right to mirror the visual presentation in the main text figures, moving from the largest observed membership declines to the smallest declines (or relative gains).

Table A.6: Impact of Changes in Dues-Paying Membership on Public Sector Earnings Growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Short-run change in dues-paying membership (2018–2019)								
% Δ Dues-Paying Membership	-0.0711 (0.055)	-0.0723 (0.055)	-0.0578 (0.057)	-0.0576 (0.058)	-0.0681 (0.055)	-0.0712 (0.058)	-0.0541 (0.054)	-0.0564 (0.058)
Wild-Bootstrap p-value	0.228	0.216	0.349	0.355	0.244	0.258	0.346	0.368
R-squared	0.00463	0.168	0.00911	0.172	0.00504	0.168	0.0106	0.172
Panel B: Medium-run change in dues-paying membership (2018–2021)								
% Δ Dues-Paying Membership	-0.0374 (0.045)	-0.0479 (0.040)	-0.0246 (0.042)	-0.0344 (0.035)	-0.0358 (0.046)	-0.0471 (0.042)	-0.0230 (0.041)	-0.0340 (0.035)
Wild-Bootstrap p-value	0.406	0.281	0.539	0.342	0.432	0.313	0.563	0.364
R-squared	0.00241	0.167	0.00722	0.171	0.00305	0.167	0.00896	0.171
Number of observations	398	398	398	398	398	398	398	398
Controls & Fixed Effects								
State Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Gov. Type Fixed Effects	No	No	Yes	Yes	No	No	Yes	Yes
Log Baseline Size (2018)	No	No	No	No	Yes	Yes	Yes	Yes

Notes: This table reports OLS regression coefficients where the dependent variable is the percentage change in annual earnings for full-time employees. Standard errors, clustered at the state level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Panel A (Short-run): Evaluates the relationship between the percentage change in dues-paying membership from 2018 to 2019 and earnings growth from 2017 to 2022.

Panel B (Medium-run): Evaluates the relationship between the percentage change in dues-paying membership from 2018 to 2021 and earnings growth from 2017 to 2022.

Wild-Bootstrap p-value: Reports p-values from a Wild Cluster Bootstrap using Webb weights and 999 replications to account for the finite number of state clusters (N=21 states).

Controls: Columns (5) through (8) include the natural log of 2018 dues-paying membership to control for baseline jurisdiction size.

Fixed Effects: State fixed effects control for time-invariant state-level characteristics. Government type fixed effects control for differences across counties, municipalities, school districts, and states.

Sample: All specifications are based on a balanced panel of 398 jurisdictions.

Across both time horizons and all 8 model specifications, the Wild-Bootstrap p-values consistently exceed 0.20, indicating no statistically significant relationship between membership changes and earnings growth.

Table A.7: Summary statistics for the percent of public-sector employees who report being union members in the CPS

Public Sector			Private Sector	
	2017	2021	2017	2021
Panel A: Levels (Percent of workers reporting union membership)				
Treated states (N=21)	48.7	47.6	8.0	7.9
Control states (N=26)	18.6	17.8	4.3	3.9
Panel B: Pre- to post- <i>Janus</i> changes (Percentage points)				
Treated states		-1.08		-0.02
Control states		-0.85		-0.37
Panel C: Estimated effects				
DiD: Public-sector treated vs. Public-sector controls		$-1.08 - (-0.85) = -0.23$		
DDD: DiD(Public) – DiD(Private)		$[-1.08 - (-0.85)] - [-0.02 - (-0.37)] = -0.58$		

Notes: Data from the CPS as compiled by Hirsch, Macpherson, and Even (2026).

Treated states are the 21 non-RTW states where *Janus* invalidated mandatory agency fees. Control states are the 26 RTW states. Colorado, Missouri, and Wisconsin are in neither the treatment nor the control group. The DiD estimate compares changes in public-sector union membership between treated and control states. The DDD estimate further nets out within-state private-sector trends. Neither estimate is economically or statistically meaningful, consistent with the regression results presented in Figure A.3.